

The Morphosyntactic Pivot Hypothesis: Hierarchical Parametric Organization of the Locus of the EPP

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1. Introduction

1.1 The problem

That T universally hosts the EPP is a standing assumption in much of the literature (Chomsky 1995, 2000), but Minimalism has already produced challenges to it from within. In pro-drop, VSO/VOS languages such as Greek, Arabic, and the Celtic languages, Alexiadou & Anagnostopoulou (1998) show that sufficiently rich verbal agreement allows the EPP to be satisfied through head-movement of V to Agr rather than through movement of an XP to Spec-TP, relocating EPP-satisfaction to Agr. A related but independent argument comes from Bobaljik & Jonas (1996) and Bobaljik & Thráinsson (1998): the availability of an additional subject-licensing position above VP is itself parametrized by the richness of verbal agreement morphology (the Rich Agreement Hypothesis), which ties functional structure to morphological paradigm richness rather than to T as a fixed, universal locus. Miyagawa (2010) pushes this further still, dividing languages into two macro-types according to whether the EPP is discharged through φ -feature agreement or through discourse-configurational (topic/focus) marking on C — a proposal §4.3 returns to directly.

Across this literature, that the EPP-host varies is already established; what remains unresolved is how to identify the pivot in a given language without circularity, and how far the variation extends beyond the T-versus-Agr and Agreement-versus-Discourse contrasts on which these proposals are built. To address this, the paper develops a non-circular, cross-diagnostic identification protocol — convergent syntactic diagnostics corroborated by independent morphological criteria — and applies it to a typologically broader sample. There, the pivot turns out to be hosted not only by T or Agr but also by Mood, Aspect, and Aspect/Evidentiality, yielding a five-way typology that §3 shows reduces to a single hierarchical two-parameter architecture, rather than remaining a language-by-language parametric list¹.

1.2 Overview of the Study

The argument proceeds in stages: developing the Biphasic Identification Protocol (§1.4); applying it systematically to identify morphosyntactic pivots in six core languages (§2); showing that the resulting variation exhibits Hierarchical Parametric Organization (§3); validating this with an 82-language statistical analysis achieving 96.60% PC1 (§3.5); and drawing out the theoretical implications (§4-5). What emerges from this diagnostic methodology is that the EPP locus is organized hierarchically rather than flatly.

1.3 The Morphosyntactic Pivot Hypothesis (MPH)

The EPP bundle is not universally tied to T but to the morphosyntactic pivot—the functional head that (i) independently meets syntactic diagnostics of structural

¹ **Supplementary Materials and Reproducibility Package**

The complete Supplementary Materials, datasets, analysis scripts, elicitation materials, and replication resources are permanently archived in the Open Science Framework (OSF):

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centrality and (ii) is corroborated by morphological prominence². That pivot projects obligatorily, values the goal's [uCase] and hosts [uEPP].

The morphosyntactic pivot is identified independently of the categories it realizes. In φ -available systems, it hosts the productive φ -feature system; in φ -unavailable systems, it may instead realize non- φ categories such as aspect, evidentiality, mood, or tense. The identity of the pivot is therefore logically distinct from its feature inventory.

1.4 Feature System and Theoretical Background

The analysis assumes standard Minimalist feature theory (Chomsky 2000, 2001, 2007; Pesetsky & Torrego 2007). Functional heads bear features that drive syntactic operations and must satisfy interface conditions. We adopt the following feature typology:

Feature interpretability:

- **[iF] (interpretable features)**: Semantically contentful; persist to LF and PF
- **[uF] (uninterpretable features)**: Formal; must be valued/deleted before transfer to interfaces
- **φ -Availability**: the availability of φ -features on the pivot. This property, together with three further diagnostics introduced in §3.5, constitutes empirical evidence for a language's φ -state (defined formally in §3.1).

Relevant features:

- **[uEPP]**: Uninterpretable EPP feature requiring satisfaction via movement to the head's specifier
- **[iCase]**: Interpretable case feature on functional heads (assigns case to DPs)
- **[uCase]**: Uninterpretable Case feature on DPs (requires valuation)
- **[i φ]**: Interpretable φ -features (person, number, gender/noun class); persist to interfaces
- **[u φ]**: Uninterpretable φ -features; require valuation and deletion
- **[n φ]**: Absence of φ -features on the pivot

Key principle: Interpretable features persist through the derivation to both LF (semantic interpretation) and PF (morphological realization). [u φ], being 'defects' must be eliminated (via valuation and deletion) before transfer to avoid interface crashes (Full Interpretation; Chomsky 1995).

φ -features and case-based intervention: The presence of φ -features on a functional head does not guarantee surface agreement morphology. In ergative systems, ergative case on the external argument blocks φ -Agree between the pivot and the subject DP (Bobaljik 1993, Laka 1993, Woolford 1997, Anand & Nevins 2006). When the pivot bears [i φ] but the goal bears ergative case, the pivot's [i φ] cannot establish matching relations with the goal's [i φ]. However, the subject DP's φ -features remain recoverable despite no matching with the pivot. If a nominative internal argument is accessible, the pivot may agree with it instead, yielding object agreement.

Relevance to pivot identification:

The pivot head hosts [iCase] and [uEPP] but varies cross-linguistically in its φ -specification. This variation correlates with morphological patterns and subject-licensing properties:

- Pivots with [u φ] (e.g., English T): impoverished agreement, no pro-drop
- Pivots with [i φ] (e.g., Swahili Agr, Hindi-Urdu Asp): rich agreement, pro-drop possible

² I apply Bybee's (1985) diagnostics, i.e., (obligatoriness, relevance, paradigmatic richness, generality) to determine morphological prominence for every language in the sample.

- Pivots with [n ϕ] (e.g., Balti Asp/Evid): no agreement, discourse/topic based pro-drop

This featural variation, its formal representation, and its interface consequences will be developed in detail in Section 3.

1.5 A Biphasic Diagnostic Protocol for Pivot Identification

Moving from the theoretical construct of the morphosyntactic pivot to its empirical identification requires a method that insulates structural diagnosis from morphological preconceptions. I therefore propose a Biphasic Diagnostic Protocol, which establishes pivothood as the point of convergence between a head's clause-delimiting syntactic properties and its featural prominence in the morphological paradigm.

Both the phases are essentially independent of each other. A functional head must pass both independent phases to be formally designated as the morphosyntactic pivot.

The independence of Phase 1 and Phase 2 is demonstrated by languages like Finnish which show complete syntactic clustering (Morph_ProDrop = 0.2, No_Expletive = 0.2) identical to Type 2 languages like English but possesses rich paradigm (Rich_Agreement 1) like Type 3 languages, receiving Phi_Availability = 0.2 (uninterpretable ϕ -features) rather than 1.0 (interpretable ϕ -features). This distinction reflects Phase 1's independent assessment of whether ϕ -features on the morphosyntactic pivot are interpretable [i ϕ] (carrying semantic content to LF, as in Italian/Spanish) or uninterpretable [u ϕ] (formal requirements deleted before LF, as in English/Finnish). The fact that languages with identical Phase 2 profiles receive different Phase 1 classifications demonstrates that the two phases assess distinct empirical domains, making circular coding impossible.

1.5.1 The Structural Autonomy Diagnostics

- **Clause-boundary diagnostics**
 - *Coordination (ConjP)*: which projection can form independent conjuncts?
 - *Complementizer selection*: which projection is selected by complementizers?
 - *Ellipsis/fragment diagnostics*: which projection survives in fragmentary answers?
- **Movement diagnostics**
 - *Subject landing site*: adverb ordering, binding, and surface subject position.
 - *Extraction/intervention*: which projections intervene in wh-movement / topicalization?
 - *Freezing effects*: whether extraction out of a moved subject is degraded.

Nomination Rule (operational). A projection is nominated as the *syntactic candidate pivot* when it clearly passes at least two clause-boundary diagnostics *and* at least two movement diagnostics. This binary, replicable rule reduces subjectivity and makes identification falsifiable (see Appendix A.2)

1.5.2 The Morphological Prominence Diagnostics

Once syntax nominates a candidate, I apply morphological diagnostics (Bybee 1985) as independent corroboration:

- **Obligatoriness**: the marker projects in most finite clauses (operational threshold: $\geq 90\%$ of attested finite clauses in naturalistic data or unanimous native-judgment).
- **Relevance (functional tie)**: the morphology is functionally tied to clause-level interpretations (tense/aspect/mood/evidentiality/agreement) and argument licensing.

- **Paradigmatic richness:** the morphosyntactic category displays multiple contrastive values (operational threshold: at least three distinct values relevant to clause-typing or φ -expression).
- **Generality/Productivity:** the morphology occurs across verb classes and clause types (operational threshold: productive for $\geq 80\%$ of attested verb types in a sample).

If morphology corroborates the syntactic candidate (strong match on two or more Bybee criteria), we accept that head as the morphosyntactic pivot. If there is divergence, the mismatch is reported and analysed (e.g., erosion, fusion, mixed pivot systems). Importantly, the morphosyntactic pivot is identified before parameter assignment. The subsequent parameters characterize the already identified pivot rather than determining its identity (see Appendix A.3).

1.6 Research Methodology

This study integrates diagnostic protocols with quantitative validation to test the Morphosyntactic Pivot Hypothesis on six core languages from different families/subfamilies. To demonstrate the morphosyntactic consequence of the MPH, i.e., Hierarchical Parametric Organization in the locus of the EPP host, the sample is extended to eighty-two genealogically unrelated languages including the six core languages.

1.6.1 Sample Selection

The core sample comprises six languages from distinct families/sub-families, i.e., English, Hindi-Urdu, Swahili, Italian, Balti, and Warlpiri, representing five continents (Hindi and Urdu, though drawn from a shared speaker sample, are coded as two separate dataset entries per standard typological practice, yielding $N=7$ in the quantitative analyses of §3.5 and Appendix D). Five languages were investigated through primary data collection ($N=110$ speakers total); Warlpiri analysis synthesizes published descriptions. Systematic expansion with seventy-five additional languages selected for genealogical diversity and typological coverage, tests the hierarchical parametric structure. The design of the study helps in creating genealogical diversity with analytical tractability (see Appendix B & C).

1.6.2 Theoretical Integration

The empirical results obtained through the application of the Biphasic Diagnostic Protocol over the primary data from the five core languages (§2) inform the theoretical development (§3), which proposes a Hierarchical Parametric Organization with two binary parameters hierarchically arranged in a logical-dependency relation:

Level 1: φ -Availability Parameter [$\pm\varphi$]

Does the empirically identified pivot bear φ -features at all?

- [$-\varphi$] $\rightarrow$ Type 1 ($n\varphi$): No φ -features (e.g., Balti) $\rightarrow$ Type 1 ('0.1' at Phi_Availability)
- [$+\varphi$] $\rightarrow$ Proceed to Level 2

Level 2: φ -Interpretability Parameter [$\pm\text{interp}$] (only if [$+\varphi$])

Are the φ -features on the empirically identified pivot interpretable at LF?

- [$-\text{interp}$] $\rightarrow$ Type 2 ($[u\varphi]$): Uninterpretable φ (e.g., English) $\rightarrow$ 0.2 Phi_Availability
- [$+\text{interp}$] $\rightarrow$ Type 3 ($[i\varphi]$): Interpretable φ (e.g., Hindi-Urdu, Swahili, Italian, Warlpiri) $\rightarrow$ 1.0 Phi_Availability

In this context, logical-dependency relation is defined as the one in which the value of the higher parameter determines the value of the lower parameter. This hierarchical dependency logically reduces the parametric space from $2^2 = 4$ combinations to 3 attested types, as [$-\varphi, \pm\text{interp}$] is incoherent (absent features cannot be interpretable or

uninterpretable). This feature composition indicates an ontological/computational asymmetry, not a stipulated convention — existence necessarily precedes computation over what exists (§3.3).

1.6.3 Statistical Validation of Parametric Dependencies

To test whether this hierarchical architecture correctly captures the organization of morphosyntactic variance, I employ Principal Component Analysis (§3.5) on four independently documented phenomena³: (i) ϕ -Availability, (ii) agreement morphology richness, (iii) morphological pro-drop licensing, (iv) and expletive absence. These variables operationalize the properties predicted to cluster under the ϕ -Availability Parameter and the ϕ -Interpretability Parameter.

To avoid any possibility of circularity, it is crucial that the four variables remain operationally independent. Each variable targets a distinct empirical domain: *Phi_Availability* is diagnosed via bound-variable licensing and coordinated-subject agreement resolution (Appendix F §2.2), neither of which references null-argument licensing, agreement richness, or expletive status. In the feature matrix, *Rich_Agreement* is concerned with measuring the morphological exponence of person/number distinctions irrespective of their interpretive status. While rich agreement is precisely a morphological feature, *Morph_ProDrop* and *No_Expletive* are diagnosed syntactically via separate elicitation batteries (Appendix D) targeting null-subject licensing and expletive-construction acceptability by the native speakers of five core languages. This operational independence may diverge empirically (e.g., Type 2b languages with ϕ -availability but rich verb paradigms, or TAM-rich systems with rich paradigms but no ϕ -features; see Appendix B.2), and ensures that any observed correlation in §3.5 reflects genuine convergence on the shared latent ϕ -state (§3.1) rather than shared operational basis.

Theoretical Predictions:

If the Hierarchical Parametric Organization is correct, the statistical analysis should reveal:

P1: One dominant dimensional structure (PC1 > 70%). If the ϕ -state of a pivot determines the morphosyntactic structure, one dimension should serve as the primary organizing dimension with other dimensions subservient to it. This single dominant dimensional structure tests whether the properties traditionally treated as independent (agreement, pro-drop, expletives), are the computational reflexes of a single latent dimension.

P2: High internal consistency ($\alpha > 0.80$). The value of the four diagnostic variables should form a coherent group as the values of these variables are not determined in isolation. The interrelationship between these variables confirm that they measure a unified construct rather than orthogonal phenomena.

P3: Non-monotonic behavioural clustering tracking ϕ -Interpretability, not mere ϕ -presence. Type 1 ($n\phi$) and Type 3 ($i\phi$) both permit null subjects and disallow expletives — Type 1 via discourse-pragmatic licensing in the total absence of a ϕ -probe, Type 3 via morphological licensing through an interpretable ϕ -probe. Type 2 (ϕ -present but uninterpretable) is the sole type that blocks null-subject licensing and requires expletives: it possesses a ϕ -probe, unlike Type 1, but that probe delivers no interpretable content capable of licensing recovery of a null

³ These phenomena have independent research histories: Corbett (2006) on agreement, Rizzi (1982) on pro-drop, Chomsky (1981) on expletives, and Chomsky (1995) on ϕ -features. Their high correlation ($r = 0.78$ – 1.00) is an empirical discovery demonstrating convergent validity.

subject, unlike Type 3. The predicted pattern is therefore U-shaped in φ -Availability rather than monotonic: Type 2 is not an intermediate point between Type 1 and Type 3, but the qualitatively most restrictive type of the three, precisely because it combines the syntactic cost of a φ -probe with none of its interpretive benefit — the asymmetry developed in §3.3.

P4: A flat, non-hierarchical alternative in which φ -Availability alone determined pro-drop and expletive behavior would instead predict a monotonic gradient (Type 1 $\leq$ Type 2 $\leq$ Type 3), with Type 2 patterning as an intermediate case rather than the outlier. Finding the U-shaped pattern instead of the monotonic one is therefore a distinguishing empirical test of the hierarchical model, confirmed by the type profiles in §2.7 and Table 4 (§3.5).

The statistical validation directly supports the logical-dependency between φ -Availability and φ -Interpretability. Had the two parameters been independent of each other (flat architecture), there should have been four logically possible combinations, distributed across two comparably weighted dimensions, yielding multidimensional structure (PC1 $\approx$ 35–50%). However, if instead φ -Interpretability is strictly dependent on φ -Availability — Level 2 is defined only where Level 1 = [+ φ]. It leads to the valid state space collapse from four cells to three, and those three states fall on a single ordinal scale (0.1 < 0.2 < 1.0) rather than an unordered set. This predicts unidimensional structure (PC1 > 70%) with three discrete clusters positioned along that scale, not two. The concentration of variance in PC1 (96.60%) and PC2 (3.40%), together with the three empirically distinct types confirmed in §3.5 (Table 4) and the U-shaped pro-drop/expletive pattern above, confirms this, validating the hierarchical dependency empirically.

1.6.4 Independent Corpus Validation via Universal Dependencies

To further guard against circularity, the hand-coded Phi_Availability values were tested against an entirely independent source: raw morphological data extracted from the Universal Dependencies treebank corpus (v2.17, 339 treebanks, 186 languages), a resource built by hundreds of annotators across two decades with no relationship to MPH. A corpus-derived φ -index — a mechanical, zero-learned-parameter count of Person/Number/Gender feature co-occurrence on finite verbs — was computed for 80 of the 82 sample languages (excluded: Swahili, Balti, validated instead via primary elicitation data per §1.6.1) and compared against the gold Phi_Availability coding using a fixed, theory-derived threshold set a priori, never fit to the gold labels. Because the φ -index measures only morphological richness while MPH's diagnostics for interpretability are syntactic (pro-drop licensing, expletive obligatoriness), the theory's own architecture predicts, in advance of running this comparison, exactly two failure modes for the proxy and no others. First, languages whose φ -features remain syntactically available, but whose agreement morphology has eroded should appear as false negatives at the availability level — the proxy will read them as phi-less despite phi being syntactically active. Second, languages whose φ -features have become uninterpretable while their agreement morphology survives diachronically (the Type 2b profile) should appear as false positives at the interpretability level — the proxy will read them as interpretable because it cannot see that morphology has decoupled from interpretability. No other failure pattern is predicted: the proxy should not misclassify languages at random, nor fail on languages where morphology and the syntactic diagnostics agree.

Results: the φ -index recovers φ -availability (0.1 vs. {0.2, 1.0}) at 97.5% accuracy (AUC=0.998, N=80) and φ -interpretability (0.2 vs. 1.0) at 95.0% accuracy (AUC=0.966,

N=80). Critically, the four residual misclassifications fall exactly into the two failure modes predicted above, and no others — not one undifferentiated group of noise, but a structured match to a prediction stated before the comparison was run. Norwegian and Swedish are false negatives at the ϕ -availability level (predicted 0.1/phi-less, gold says 0.2): both have completely lost verbal person-marking morphology, identical in surface appearance to genuinely phi-less languages like Chinese — yet the syntactic diagnostics reveal phi is nonetheless present, evidenced by the same two hallmarks that identify German, English, and Dutch as 0.2 languages: the absence of morphological pro-drop and the obligatoriness of expletives. Icelandic, German, and Finnish constitute false positives at the interpretability level (predicted 1.0/interpretable, gold says 0.2) and represent the opposite profile: in sharp contrast to Norwegian and Swedish, these languages retain genuinely rich verbal agreement (Rich_Agreement=1). However, the richness of morphology does not entail interpretability, precisely the central empirical claim of Type 2b. That the ϕ -index, built purely from morphological richness, cannot distinguish either case from Type 3 is expected: it has no access to the pro-drop and expletive diagnostics that correctly identify all five languages as 0.2 regardless of whether their agreement morphology is rich or entirely absent. This constitutes a genuine, independent-data test of MPH's core prediction — one that could have failed, in a specific and pre-stated way, and did not fail outside that pre-stated pattern.

1.6.5 Data Availability, Transparency and Reproducibility

The study is designed for full reproducibility. Elicitation materials, raw acceptability-judgment data, morphological codings, the complete eighty-two-language dataset, the Universal Dependencies validation script and treebank-derived ϕ -index outputs (§1.6.4), and a single R script reproducing every figure and statistic reported are provided in the online Supplementary Materials. All raw datasets, analysis scripts, and reproducibility materials are permanently and publicly available via the Open Science Framework: <https://doi.org/10.17605/OSF.IO/K95P2>, in accordance with journal policy and ethical requirements for anonymized participant data.

Limitations & pre-registered mitigations

- **Sampling limits:** for smaller-language corpora, the obligatoriness threshold is lowered and impacts discussed; sensitivity analyses are reported.
- **Dialectal variation:** where dialect splits are known (e.g., Balti varieties), dialect-specific responses are reported, or collapsed only after verifying homogeneity.
- **Subjectivity in diagnostics:** operational thresholds and dual-coder checks reduce the threat of subjective labelling.

Future evidence may require refinement of individual diagnostics without necessarily affecting the underlying theoretical architecture.

1.7 Empirical Scope

The Biphase Identification Protocol is applied to the six core languages, with syntactic diagnostics identifying the pivot and morphological diagnostics providing corroborating evidence. Five core languages underwent extensive syntactic testing with native-speaker judgments (N=110); the sixth was analyzed via published description. The seventy-five outgroup languages were not tested under the core protocol; their morphosyntactic profiles were developed from existing descriptive and theoretical literature, cross-checked against Universal Dependencies treebank data where available (§1.6.4).

The Biphase Diagnostic Protocol is applied to all six core languages. The diagnostics are uniformly applied for all the core languages. In each case, the same sequence of procedure is applied over each core language (§2). Each application follows the same template: (i) syntactic battery results, (ii) morphological corroboration by Bybee

criteria, (iii) and the verdict on pivothood. I provide a comparative synthesis and the theory (§3) (full diagnostic detail is in Appendix E).

1.8 Consequences

The Biphasic Diagnostic Protocol is designed to reduce the risk of circularity in the process of pivot identification. The pivot identification across languages leads to a predictable and principled typology. This protocol ensures that the core commitments of the present study are straightforward, the thresholds are made explicit, coding is made transparent, and the analysis is reproducible. The evidence from the UD treebanks provides independent evidence in favor of the statistical findings.

The Balti case demonstrates an under-documented third parametric type: pivots without any φ -specification ($n\varphi$). Native-speaker judgments (N=30, Skardu fieldwork) uniformly report no person/number distinctions on aspectual/evidential morphology; the evidential system distinguishes egophoric, inferential, reportative, and visible-evidence categories, but these encode epistemic, not agreement, distinctions. This illustrates a system where case licensing and EPP satisfaction occur independently of φ -Agree (§2.5, Appendix E.5). The absence of productive φ -features does not imply the absence of a morphosyntactic pivot. In φ -unavailable systems, the pivot may instead realize non- φ categories such as evidentiality or aspect.

1.9.1 Theoretical Reformulation of the EPP: The primary contribution here is a path toward systematically decoupling the EPP from Tense. I show that the requirement for a structural anchor is satisfied by whichever projection is morphosyntactically most central in a given language, namely the pivot. This is why Hindi-Urdu shows EPP effects such as subject licensing and pro-drop patterns anchored in Aspect rather than Tense, and why Warlpiri satisfies its EPP through head movement of its agreement complex instead.

1.9.2 A fully operationalized falsifiable minimalist parameter system: I develop a compact set of hierarchically organized φ -related parameters whose values come exclusively from independently testable syntactic diagnostics rather than from descriptive generalizations or post hoc typologizing. As proposed by present study, the Biphasic diagnostic protocol runs from a Syntax Phase to a Morphology Phase. Each of the phase is defined with explicit criteria, threshold conditions, and falsifiable tests. The ordering of syntax and morphology here matters as it is evident from German and Icelandic retaining rich verbal agreement morphology yet patterning as $[u\varphi]$ rather than $[i\varphi]$. Thus, if a protocol infers interpretability from morphological richness alone, it will misclassify these languages. The Syntax Phase determines Φ _Availability independently of the Morphology Phase. That is why the protocol correctly separates these languages from Type 3 despite their superficially rich paradigms of these languages. This result is a consequence of the diagnostic architecture itself, instead of needing a language-by-language exception list.

This diagnostic architecture provides a maximally objective, reproducible, and coder-independent parameter assignments. Even more important consequence of the diagnostic structure of the present study is that it offers such a theory as is falsifiable in the classic Popperian sense (Popper 1959). The framework does not simply show what is possible; it also shows what is impossible, meeting Popper's criteria of classic falsifiability.

The MPH framework concretely predicts that Φ _Availability, Morph_ProDrop, and No_Expletive derive from a single latent dimension i.e., φ -state. Therefore, they are predicted to converge deterministically. Thus, the theory may be falsified by any language that demonstrates that these three variables do not converge. However,

Rich_Agreement is not part of this inviolable core. This exclusion is principled rather than post hoc: interpretability of the φ -features is established through syntactic diagnostics, not morphological richness. That is why agreement can persist even after interpretability is lost (Type 2b) or erode while phi remains syntactically active. This architecture pre-specifies, not merely accommodates, exactly where a morphology-only measure should fail — a prediction tested and confirmed in §1.6.4

1.9.3 A two-level architecture with internal logical dependence: φ -features are organized into two levels: φ -Availability, assessed in narrow syntax, and φ -Interpretability at LF, assessed at the interface. Interpretability is defined only if Availability holds. It is due to this logical-dependency instead of any stipulation that the logical space is reduced from four combinations to three attested macro-types, since $[-\varphi, \pm\text{interp}]$ is illogical as the absent features can neither be interpretable nor uninterpretable. The exclusion of the fourth macro-type is a consequence of the derivationally constrained, minimal typology of the φ -systems.

1.9.4 Identification of an under-recognized configuration in existing typological treatments: the framework predicts a specific type of language — the one lacking φ -features in the narrow syntax but with nominal interpretability at the LF. Based on my original fieldwork in the Skardu region, Pakistan, I demonstrate that Balti is exactly that configuration which the framework predicts. It constitutes a striking case in which a theoretically necessary type turns out to be empirically discoverable and statistically validated, further validated by the UD trees. Owing to documenting this relatively under-explored configuration, the study provides statistically validated insight into how subjects, pivots, agreement, and clause structure relate to one another and form a coherent syndrome.

1.9.5 A Unifying Theoretical Framework: The study demonstrates that Phi_Availability, Rich_Agreement, Morph_ProDrop, and No_Expletive form a coherent syndrome. Instead of being simply correlated properties, they are statistically demonstrated to be the computationally linked reflexes of a single underlying dimension, i.e., φ -state. Each of these variables, being a predictable consequence of a pivot having a categorical φ -state, offers a principled, causal explanation for why these properties tend to co-vary cross-linguistically.

Principal Component Analysis (PCA) is employed on a sample of eighty-two-languages to provide systematic statistical evidence of the φ -state constituting the primary dimension. It is confirmed by the PCA where PC1 explains 96.60% morphosyntactic variance. Phi_Availability is tied as the strongest-loading indicator, along with Morph_ProDrop and No_Expletive (25.61% each). These four variable together account for the morphosyntactic variation across languages with PC1 and PC2 together accounting for 100% of the structured morphosyntactic variance. This internal statistical structure is independently corroborated by an external test: φ -Availability values derived mechanically from raw Universal Dependencies treebank morphology. These treebanks were built without any reference to MPH — recover the hand-coded classifications at 95–97.5% accuracy (§1.6.4). Every residual discrepancy is independently explained, instead of being explained away, by the theory's own finer distinctions rather than left as noise. The convergence of the theoretical predictions, statistical structure, and independent UD corpus validation provides strong empirical support for the MPH framework.

Taken together, these contributions establish the Morphosyntactic Pivot Hypothesis (MPH) as:

- an operational, testable, statistically validated, and reproducible minimalist model, independently corroborated by corpus data external to the theory,
- consistent with a configuration predicted by the theory but not clearly documented before,
- a demonstration of a compact parameter set accounting for the full extent (100%) of observed structured variance

2. Summary of Cross-Linguistic Findings

This section deal with the implementation of the Biphasic Diagnostic Protocol— syntactic nomination followed by morphological corroboration— on the core six languages. It summarizes the outcomes of the implementation of this biphasic protocol. All the diagnostic inventories, sample sentences, glosses, judgements and quantitative results are provided in Appendix E (§§1–6). This section highlights only the convergent empirical evidence identifying the morphosyntactic pivot in each of the core six systems.

2.1 English: T as the Morphosyntactic Pivot

Syntactic identification.

In English, clause-boundary diagnostics consistently point to T(ense) as the clause-defining head. The coordination test reveals that TP is the natural unit: the data demonstrate that finite clauses coordinate only if both conjuncts bear tense (*John left and Mary arrived*); aspectual or agreement morphology alone does not license coordination. The complementizer selection test, too, highlights the same pattern. The native speaker judgements show that the complementizer *that* uniformly selects a TP as its complement, yielding embedded finite clauses with tense specification (*I know that John left*). Third, ellipsis patterns show that auxiliaries associated with T survive as remnants in fragment answers (*Did John leave? – Yes, he did*), indicating that TP material anchors clause-sized ellipsis (see Appendix E.1.1.1)

Movement diagnostics provides further support to T as the pivot. As demonstrated by the adverb ordering and binding asymmetries (subjects surface above VP-internal adverbs and their binding domains), subjects are observed obligatorily raising to Spec-TP. Extraction asymmetries also implicate T as the morphosyntactic pivot: subject movement interacts with Wh- and topicalization processes in ways which are fully consistent with an intermediate stop in Spec-TP. Freezing effects too confirm that extraction from within a subject that has raised to Spec-TP is degraded; it is strictly in line with predictions if T hosts the EPP feature. All the syntactic diagnostic tests uniformly point to T as the syntactic candidate pivot in English (see Appendix E.1.1.2).

Morphological corroboration.

Syntactic nomination of T as the morphosyntactic pivot in English is further corroborated by the morphological diagnostics. Bybee's (1985) criteria provide partial but sufficient support to corroborate the syntactic nomination of T as the pivot. T meets Bybee's criterion of obligatoriness as tense marking in English is obligatory in finite clauses. Every finite verb form obligatorily requires tense marking (past vs non-past), which is typically realized on the auxiliary or main verb. English has an impoverished verbal morphology with only two tensed forms. That is why paradigmatic richness of verb is comparatively reduced with the contrasts being largely restricted to past vs. non-past and the 3sg present marker -s. However, these distinctions are stable. Productivity is high: tense marking extends across the verbal lexicon and applies to novel forms without restriction (see Appendix E.1.2).

Verdict

English thus illustrates a system where syntactic diagnostics and morphological criteria

both converge on T as the morphosyntactic pivot, even though the tense paradigm itself is relatively impoverished (see Appendix E.1.3)

2.2 Hindi-Urdu: Asp as the Morphosyntactic Pivot

Syntactic identification.

Unlike T in English, clause-boundary diagnostics clearly point to Asp as the morphosyntactic pivot in Hindi-Urdu. The judgement data show that coordination works only at the level of AspP. A Hindi-Urdu clause is possible without an overt tense auxiliary (*Abdullah-ne roti khā-ī aur kofī pī-ī* ‘Abdullah ate bread and drank coffee’). The lack of AspP in Hindi-Urdu leads to ill-formedness. Thus, a Hindi-Urdu sentence shall accommodate only the clauses with obligatory AspP. The complementizer selection too supports this view: the complementizer *ki* (‘that’) can embed clauses that contain aspectual marking but lack tense morphology (*Main chāhtā hū̃ [ki Abdullah roti khā-ī]* ‘I want that Abdullah eat bread’). Like coordination and complementizer selection, ellipsis provide further empirical support and confirms Asp’s clausal status. The data consistently demonstrate that, in fragment answers, it is aspectual morphology that survives (*Khā-ī* ‘ate’), while tense morphology may be absent altogether (see Appendix E.2.1.1).

Movement diagnostics also lead to the same conclusion. Subjects are observed to surface in Spec-AspP, instead of Spec-TP in Hindi-Urdu. In perfective clauses, ergative subjects consistently precede aspectual morphology, demonstrating that the subjects move to a position licensed by Asp instead of T. Wh-subjects also pass through AspP en route to CP. The adverb placement and intervention effects reveal such an intermediate landing site in Spec-AspP. Likewise, freezing effects become noticeable as the extraction from such a subject as occupies Spec-AspP leads to degradation. This finding is fully consonant with Asp hosting the EPP-related subject position. All the syntactic diagnostics unanimously nominate Asp as the syntactic candidate pivot (see Appendix E.2.1.2).

Morphological corroboration.

The morphological evidence obtained through the application of Bybee’s criteria corroborates the syntactic identification of Asp as the pivot. In finite Hindi-Urdu clauses, there must be obligatory aspectual morphology. In the absence of aspectual morphology, the obligatory specification for perfective, progressive, or habitual aspect are missing, hence, ill-formedness. Unlike English impoverished verbal morphology, Hindi-Urdu verbal morphology is paradigmatically rich. It marks not only aspectual contrasts but also gender and number agreement on participles. Critically, Asp bears [iφ] in both imperfective and perfective contexts — the same functional head. But ergativity in perfective clauses blocks subject-agreement and instead permitting object-agreement to surface (see Appendix E.2.2).

Verdict.

The Biphasic Diagnostic Protocol unambiguously establishes that the syntactic diagnostics and morphological criteria together highlight Asp as the morphosyntactic pivot in Hindi-Urdu. The identification of Asp as the morphosyntactic pivot confirms that clause-defining functions can be anchored in heads other than T. This case also shows how assessing syntax before morphology helps reduce the risk of circularity (see Appendix E.2.3).

2.3 Swahili: Agr as the Morphosyntactic Pivot

Syntactic identification.

The application of independent syntactic diagnostics in Swahili points to yet another functional head as the morphosyntactic pivot, i.e., Agr (agreement morphology on the

verb) as the clausal pivot. The clause-boundary diagnostics show that finite clauses require subject agreement prefixes. Swahili conjuncts cannot coordinate two finite clauses unless both carry agreement. Likewise, embedded clauses introduced by complementizers still obligatorily display agreement. Ellipsis provides further support to this position: the agreement prefix suffices to license a null subject (*Ali-enda? – Ndiyo, a-li-enda* ‘Did he go? – Yes, he went’) in fragment answers. Thus, instead of T or Asp, agreement morphology (Agr) acts as the defining marker of clausehood in Swahili. The clause-boundary diagnostics unanimously point to Agr as the morphosyntactic pivot (see Appendix E.3.1.1).

Along with the clause-boundary diagnostics, movement diagnostics also provide further corroborative support to this conclusion. The landing site of subject and its licensing are closely tied to Agr in Swahili. That is precisely why pro-drop is possible only when the subject agreement prefix is present in the clause because Agr can authentically recover the subject. The asymmetries observed during the Wh-extraction and topicalization interact with agreement in ways that show Agr is an intervener in movement paths. Likewise, freezing effects are also consistent with this analysis because once a subject is licensed via Agr, internal extraction from it is degraded. Thus, both the clause-boundary and movement diagnostics collectively nominate Agr as the syntactic candidate pivot (see Appendix E.3.1.2).

Morphological corroboration.

This syntactic nomination of Agr as the pivot is further supported by the morphological evidence. Agreement morphology is obligatory in all finite clauses, with no exceptions. It is paradigmatically rich, marking person/number distinctions alongside an extensive noun-class system. It is also systematically productive, attaching to virtually all verbs in the language and extending across clause types. Unlike tense or aspect markers, agreement morphology alone suffices to signal finiteness (see Appendix E.3.2).

Verdict.

Syntactic and morphological evidence together identify Agr as the morphosyntactic pivot in Swahili, consistent with the pro-drop pattern. Subject licensing here is rooted in agreement rather than in tense or aspect (see Appendix E.3.3).

2.4 Italian: Mood as the Morphosyntactic Pivot

Italian serves as the test case for the Morphosyntactic Pivot Hypothesis because instead of presenting a terminal case, it represents a stage in the evolution of clause structure over the period. Whereas Latin maintained a strict division between Tense and Mood morphology, it has developed over a period a high degree of morphological fusion between the two while retaining syntactic independence.

Syntactic Identification

All the standard clause-boundary and movement diagnostics including coordination, complementizer selection, ellipsis, subject-position tests, extraction, and freezing effects—uniformly point to Mood as the locus of clausal organization. Clauses marked for Mood are observed to behave as complete finite units, and subjects are licensed in the structural domain associated with Mood rather than Tense.

As Italian is a complex case of fusion, I applied a series of seven additional diagnostics with the focus on the phenomena that are particularly sensitive to clause-internal feature hierarchies: polarity licensing, temporal-dependency patterns, scope relations, particle distribution, ellipsis asymmetries, subject-position contrasts, and, most decisively, the clitic-placement test. Together these diagnostics provide converging evidence that Mood, not Tense, defines the finite clause in Italian. Across all tests, mood variation categorically affects grammaticality (100% → 0% acceptance), whereas tense

variation within a given mood produces no syntactic contrast (see Appendix E.4.1.1 & E.4.1.2).

Morphological Corroboration

Notwithstanding the observed pervasive fusion of Mood and Tense morphology, the morphological data corroborate the syntactic nomination of Mood as the syntactic pivot. Mood specification is obligatorily marked in all finite clauses because every finite verb overtly marks one of the major moods (indicative, subjunctive, conditional, or imperative), and these paradigms are fully productive across verb classes. This obligatoriness is observable in contexts where partial separation is observable. In imperatives (Mood without Tense) and subjunctives (Mood-determined temporal interpretation), Mood remains obligatory in three of the four clause types, whereas Tense is obligatorily realized in only one. This asymmetry confirms that the Italian clause is anchored in Mood (see Appendix E.4.2).

Verdict and Diachronic Interpretation

The syntactic nomination of Mood as the pivot through the thirteen syntactic diagnostics, including the decisive clitic-placement test, converges with the robust morphological obligatoriness of Mood as the morphosyntactic pivot in Italian. Like Swahili (Agr + [i ϕ]) and Hindi-Urdu (Asp + [i ϕ]), Italian is a pivot-switch language whose interpretable ϕ -features license pro-drop; English (T + [u ϕ]) contrasts with all three in blocking pro-drop. Synchronic analysis cannot resolve the persistent Mood–Tense fusion. It instead marks a distinct diachronic phase in the language's evolution, one where syntactic reorganization anchoring the EPP in Mood has already taken place while morphological transparency has lagged behind. The fusion reflects historical layering, in other words, rather than analytic indeterminacy, and captures the shift from the T-based clause of Latin to the Mood-based architecture of modern Italian (see Appendix E.4.3).

2.5 Balti: Asp/Evid as the Morphosyntactic Pivot

Syntactic identification

In Balti, clause-boundary diagnostics consistently nominate Asp/Evid as the pivot head without any ϕ features. Coordination demonstrates that clauses can conjoin around evidential or aspectual auxiliaries (*Na-la aasho thong-sad pa dang go-sad* ‘I saw the woman and (I) went’), even in the absence of an independent tense marker. The selection of complementizer highlights the same result. The data consistently show that Balti complementizers *kih* embed clauses whose finiteness is guaranteed by evidential auxiliaries (*Na la shespo kih kho dy aasho la hlataen yod* ‘I think that he/she is seeing the woman’). Ellipsis further confirms this role as the evidential auxiliary (*yod, yin, song, zer*) survives in fragment answers, showing that Asp/EvidP alone anchors clausehood (see Appendix E.5.1.1).

Likewise, the movement diagnostics too reinforce the identification of Asp/Evid as the morphosyntactic pivot. Subject DPs in Balti consistently surface in Spec-Asp/EvidP, not Spec-TP like English. Ergative subjects in perfective clauses obligatorily appear above the auxiliary (*Na-la aasho thong-song* ‘I saw the woman’), revealing Asp/Evid as the landing site. On their way to CP, Wh-subjects also pass through this position as shown by their placement relative to adverbs (*Su-la aasho thong-sad?* ‘Who saw the woman?’). Freezing effects arise when subjects occupy Spec-Asp/EvidP: extraction from within such subjects is degraded, consistent with the projection hosting the EPP bundle. Both the clause-boundary and movement diagnostics collectively nominate Asp/Evid as the syntactic candidate pivot (details of judgments from N=30 Balti speakers are provided in Appendix E.5.1.2).

Morphological corroboration.

This syntactic identification of Asp/Evid as the morphosyntactic pivot is further supported by the morphological evidence. The obligatoriness of Evidential and aspectual auxiliaries regardless of tense specification in all finite clauses support Asp/Evid as the morphosyntactic pivot. Obligatory but alternating aspectual/evidential auxiliaries (*yod, yin, song, zer*), make distinction between perfective, progressive, evidential, and epistemic forms. Thus, the system is paradigmatically rich yet possesses no person/number distinctions. It is also systematically productive, applying across the full verbal lexicon and extending to novel predicates. By contrast, tense markers alone are insufficient to define clausehood (see Appendix E.5.2).

Verdict

Syntactic and morphological evidence together establish Asp/Evid as the morphosyntactic pivot in Balti. This makes it typologically parallel to Hindi-Urdu. However, Balti differs in one respect: evidentiality plays a central role in defining clause structure there (see Appendix E.5.3).

2.6 Warlpiri: Agr as the Morphosyntactic Pivot

Note on data provenance: Unlike other core five languages having primary data, this analysis relies on published descriptions (Hale 1973, 1983; Simpson 1991; Legate 2002, 2008; Laughren 2002; Bovern 2014) and does not include primary fieldwork. Therefore, the claims are accordingly restricted to what can be established from the cited literature and existing corpora.

Syntactic identification.

The application of clause-boundary diagnostics on Warlpiri leads to Agr (the auxiliary clitic complex) as the clause-defining morphosyntactic pivot. The coordination test shows that clauses require the auxiliary complex as the coordinated clauses each bear auxiliaries encoding subject and object agreement (*Ngarrka-ngku ka karli jarnti-rni manu pama-rnu* 'The man is trimming a boomerang and speared it'). The coordination without auxiliaries is ungrammatical. The complementizer selection test provides further empirical support because complementizers select clauses containing the auxiliary complex, which encodes agreement and occupies the second position (Wackernagel position) in the clause. In the same way, ellipsis patterns also show that the auxiliary can survive as a minimal fragment, anchoring clause structure independently of the main verb (see Appendix E.6.1.1).

The battery of movement diagnostics supports Agr as the morphosyntactic pivot. They reveal a distinctive pattern. Despite following a radical free word order for lexical DPs, the auxiliary complex occupies a fixed second position. This auxiliary complex hosts the morphological features of subject; and instead of satisfying it through DP movement to Spec-AgrP, Agr satisfies the EPP through head-movement of the auxiliary itself to the fixed clausal second position. Extraction processes interact with the auxiliary position, showing intervention effects consistent with the auxiliary as a phase boundary. The distinctive property is that subjects (and other arguments) remain in or near vP, with agreement features checked via the auxiliary that has moved to second position. The syntactic and morphological diagnostics nominate Agr as the predicted morphosyntactic candidate pivot, though the system radically differs from other pivots in satisfying the EPP through head-movement instead of the phrasal movement to Spec-PivotP (see Appendix E.6.1.2).

On the status of second position: As the analysis claims that Warlpiri auxiliary movement to second position satisfies the EPP, it raises the question of whether second position is merely a PF (Wackernagel) requirement or a syntactic position. The evidence

from the existing corpora suggests both: second position is syntactically fixed and constitutes the locus of EPP satisfaction.

Evidence for syntactic status:

Complementizers occupy second position: The complementizer *kuja* 'that' appears in second position in embedded clauses, suggesting second position is a syntactic slot (like C° or a high functional head) rather than purely prosodic placement.

Structural fixedness: Irrespective of the prosodic or information-structural factors, auxiliaries cannot appear clause-finally or in other positions. This rigidity in the second position of auxiliary complex indicates syntactic positioning rather than variable PF adjustment.

Extraction interaction: The data show that long-distance wh-extraction shows that movement interacts with auxiliary position systematically. The auxiliary complex is observed to mark phase boundaries that extraction must transit. This strongly suggests that the auxiliary occupies a syntactically active position mediating A'-dependencies.

Agreement obligatoriness: Morphologically encoded ϕ -features of subject and object on the auxiliary are obligatorily realized and cannot be omitted even if the subject and object DPs are overtly present. This is evident from this syntactic obligatoriness that the auxiliary position itself is syntactically required, not merely phonologically convenient.

The analysis presented here views the fixed second position as a syntactic projection where the auxiliary (Agr) head moves to satisfy [uEPP]. This movement creates an obligatory functional projection in a structurally fixed location. This runs parallel to DP movement to Spec-TP in configurational languages but realized through head-movement rather than phrasal movement. Thus, Warlpiri demonstrates that the EPP requirement (obligatory structural anchoring) can be satisfied through different movement strategies across languages (see Appendix E.6.1.1 and E.6.1.1.2).

Morphological corroboration.

Morphological evidence converges with the syntactic identification of Agr as the morphosyntactic pivot. The existing corpora show that the auxiliary complex in Warlpiri is obligatory in all finite clauses, appearing in over 95% of matrix clauses in corpus data (Bower 2014). It exhibits a rich paradigm, creating person and number distinction for both subject and object across an extensive paradigm (1/2/3 × SG/DU/PL for subjects; similar distinctions for objects). The system is productive, applying to novel predicates, and general, extending across all verb classes without exception. By contrast, tense/aspect morphology appears on the main verb but is optional in many contexts and less paradigmatically elaborate (see Appendix E.6.2).

Verdict

Syntactic diagnostics strongly identify the auxiliary, the Agr complex, as the pivot, and morphological evidence provides robust corroboration. Warlpiri exemplifies an Agr-pivot system in which the pivot bears [i ϕ] for both subject and object, licensing pro-drop for both argument positions. Its distinctive feature is that the EPP is satisfied through pivot-head-movement to second position, rather than through DP-movement to Spec-PivotP. This gives Warlpiri a typological parallel to Swahili, another Agr-pivot language with [i ϕ] and pro-drop, but with its own, unique EPP-satisfaction strategy. What Warlpiri shows theoretically is that EPP satisfaction need not involve phrasal movement to Spec-PivotP at all. The pivot can move itself, via head-movement to second position, satisfying both the second-position requirement and the EPP's demand for an

obligatory projection. This dissociates the EPP from "subject raising" while still keeping the EPP hosted on the morphosyntactic pivot. It supports the broader MPH claim in a further way: the pivot is identified through structural centrality (second position is structurally fixed, while DPs are radically free), morphological prominence (the auxiliary is obligatory and paradigmatically rich), and EPP-hosting, and EPP satisfaction here proceeds through head-movement rather than specifier-filling. Warlpiri is thus an Agr-pivot language with [$i\varphi$], licensing robust pro-drop for both subject and object (see Appendix E.6.3).

2.7 The EPP Hosting Maxim

Across all languages investigated, the locus of the EPP-bundle—[uEPP], [i Case], and ($\pm\varphi$)—coincides with the morphosyntactic pivot identified through the Biphasic identification diagnostics. The empirical pattern generalizes as follows:

EPP-Pivot Convergence (Empirical Generalization)

The functional head that satisfies independent syntactic diagnostics of structural centrality and meets morphological criteria of prominence (obligatoriness, paradigmatic richness, relevance, and generality) is the unique host of the EPP-bundle. This head obligatorily projects, licenses case via [i Case], and anchors clause structure by valuing [uEPP]. Its φ -state is an indicator of the presence and interpretability of the φ -feature, rich agreement, morphological pro-drop and expletives.

Theoretical significance: Theories that treat syntax and morphology as loosely coupled would not predict this convergence between syntactic structural centrality and morphological prominence. It follows, however, if obligatory projection and morphological robustness target the same grammatical position — whether for learnability reasons, since surface morphology makes the pivot identifiable, or for architectural economy, since a single head then hosts all finite-clause requirements. What the EPP-Pivot Convergence captures, cross-linguistically, is that EPP-hosting is a derived property of pivothood rather than an a priori universal of T. Once the pivot is independently identified, the distribution of EPP effects follows predictably from its feature composition.

Crucially, this generalization leads to the Hierarchical Parametric Organization of the locus of the EPP itself, requiring two distinct modes of Agree. In languages whose pivots bear [$i\varphi$] or [$u\varphi$], EPP satisfaction proceeds through either Unimodal Agree (valuation of [uEPP] and [uCase] with or without φ -features) or Bimodal Agree (matching for φ -features and valuation for [uEPP] and subject's [uCase]).

Section 3 develops this theoretical consequence formally in terms of the φ -Availability and φ -Interpretability Parameters. This unified account explains cross-linguistic variation in case, agreement, pro-drop, and expletives as predictable outcomes of hierarchical structure of the EPP-hosting pivot as evidenced by the high concentration on PC1 (96.60%) and secondary structured variation on PC2 (3.40%).

3. The φ -Availability and the φ -Interpretability Parameters: Hierarchical Parametric Organization

3.1 Motivation: The Hierarchical Parametric Structure

The cross-linguistic diagnostics in Section 2 show that the morphosyntactic pivot varies systematically in its featural specification, with direct consequences for case licensing, subject realization, pro-drop, expletives, and agreement. This variation has a hierarchical structure, not a flat one: rather than selecting freely among three unrelated pivot types (0.1 $n\varphi$, 0.2 $u\varphi$, 1.0 $i\varphi$), languages make parametric choices at two distinct levels that interact predictably.

Following the terminology of latent-variable measurement theory, a language's joint setting on the φ -Availability and φ -Interpretability Parameters is its **φ -state**. φ -state is not itself a directly observed property; it is a latent grammatical parameter, instantiated in the narrow syntax as the pivot's featural specification (§3.2), detectable empirically only through four independently testable surface reflexes: the composite φ -state at LF (Phi_Availability), the richness of verbal agreement (Rich_Agreement), the availability of morphological pro-drop (Morph_ProDrop), and the licensing status of expletive subjects (No_Expletive). Each reflex is generated by φ -state but does not itself define it — φ -state is not a composite or index of the four properties, but their common cause. This reflective structure licenses treating convergence among the diagnostics (§3.5) as evidence for φ -state's existence rather than an artifact of how the diagnostics were built. That claim is tested in this study by two independent means: internal statistical structure across a genealogically diverse 82-language sample (§3.5), and external corroboration from Universal Dependencies treebank data — a corpus assembled without reference to MPH (§3.5.5). Both converge on the same latent structure, which is the strongest evidence available that φ -state is a real organizing property of grammar rather than a descriptive convenience.

A language's φ -state takes one of three values — 0.1 (φ -unavailable), 0.2 (φ -available, uninterpretable), 1.0 (φ -available, interpretable) — corresponding to Types 1–3 (§3.3). All φ -less systems uniformly lack person-number morphology on the pivot, require discourse context for null-subject licensing, lack expletives, and show aspectual or evidential morphology as the obligatory clause-defining marker. This clustering follows from hierarchical parametric interaction, not from independent parametric choices operating simultaneously.

3.2 The φ -Availability and Interpretability Parameters

3.2.1 The φ -Availability Parameter

The φ -Availability Parameter specifies whether a language activates φ -features at the pivot head at all — an ontological question, logically prior to any question of how those features are computed.

In φ -available systems, the pivot bears φ -features — interpretable or uninterpretable — and establishes φ -dependencies with nominal arguments. English, Hindi-Urdu, Swahili, Italian, and Warlpiri instantiate this setting. Finite clauses require person-number/gender/class specification somewhere in the verbal morphology, whether impoverished (English third-singular -s) or elaborate (Swahili subject prefixes distinguishing eighteen noun classes). The pivot probes to value or match subject DPs, and case licensing proceeds in tandem with φ -agreement.

In [$n\varphi$] systems, the pivot lacks φ -features entirely; clause anchoring operates through aspectual or evidential specification instead. Balti instantiates this setting: finite clauses require perfective/progressive, direct/inferential, or egophoric/non-egophoric marking, but show no person or number distinction whatsoever. The pivot projects obligatorily and licenses case, satisfying the EPP through aspectual anchoring with no φ -checking; subjects remain visible to the computational system through their own interpretable φ -features but never enter a dependency with the pivot.

φ -available systems should show person-number/gender paradigms on at least some verbal forms and permit either obligatory overt subjects or pro-drop, depending on interpretability. φ -less systems should uniformly lack such paradigms, show aspectual/evidential paradigms instead, permit pro-drop only when discourse context supplies recoverability, and show case systems varying independently of any φ -specification.

3.2.2 The ϕ -Interpretability Parameter

Once ϕ -features are activated, their interpretability determines how ϕ -relations are realized. This parameter operates only within ϕ -available systems -a computational question about the treatment of a ϕ -specification already established to exist, not about its existence.

Uninterpretable ϕ -features trigger valuation-based **Unimodal Agree**. English T bears [$u\phi$], probing for and valuing against a goal's [$i\phi$], then deleting [$u\phi$] to satisfy Full Interpretation. [$iCase$] on T values [$uCase$] on the goal as nominative in the same dependency. Because uninterpretable ϕ requires a valuator, the derivation needs an overt DP or expletive insertion when no thematic subject is available. Post-valuation deletion means ϕ -information does not persist morphologically — blocking pro-drop.

Interpretable ϕ -features trigger matching-based **Bimodal Agree**. Hindi-Urdu, Swahili, Italian, and Warlpiri instantiate this setting: pivot and argument both bear semantically contentful ϕ that must be compatible rather than one valuing the other. Because both feature sets are interpretable, they persist to both LF and PF; at PF they surface as agreement morphology recoverable even when the argument is phonologically null, making pro-drop available.

A critical refinement, established empirically in §3.5.3 and §3.8: It needs to be noted here, established empirically in §3.5.3 and §3.8, that the "cascade of consequences" from interpretability is not uniform across all four reflexes. All the four diagnostics including Phi_Availability, Morph_ProDrop, and No_Expletive except Rich_Agreement deterministically track interpretability of pivot's ϕ -features at $r=1.000$ across the full sample of eighty-two languages from twenty-one families.

Rich_Agreement does not because it can survive diachronically after a pivot's ϕ -features have already become uninterpretable. This dissociation of Rich_Agreement from the other three diagnostics is exactly what produces the Type 2b languages such as German, Icelandic, Faroese, Finnish). The theoretical claim, therefore, needs to be stated precisely that interpretability determines subject-licensing behavior categorically, but it determines agreement morphology only as a tendency.

3.3 The Hierarchical Parametric Organization

The binary ϕ -parameters stand in a logical-dependency relation. In this relation, the interpretability of a pivot's ϕ -features can be defined only when the ϕ -features are available at all. Instead of merely being a technical convenience, this dependency between availability and interpretability reflects a real asymmetry in kind between the two parameters, i.e., existence and computation of what exists. ϕ -Availability entails existence and is, therefore, termed an ontological parameter which asks whether ϕ -features exist in the derivation at all, i.e., whether the pivot bears any ϕ -specification for the computational system to operate on. In sharp contrast, ϕ -Interpretability is a computational parameter. It is concerned with asking whether the already-existing ϕ -specification is valued and deleted, under the Unimodal Agree, or is matched and retained to LF, under the Bimodal Agree. For the computation of a feature specification, the feature itself must be present in the derivation. This asymmetry between availability and interpretability is a logical necessity since existence is a precondition for computation, and no procedure can compute the interpretability of a feature that is not there. Thus, the hierarchical dependency does not follow from a stipulated ordering chosen merely to reduce four logical cells to three. Instead, it follows from the entailment that computation presupposes an object to compute over. ϕ -availability is ontologically and logically prior, not just prior by convention. In sharp contrast, ϕ -interpretability is secondary, computational refinement that operates only within ϕ -

available systems. This logical-dependency relation leads to three macro-types, not four, since $[-\phi, \pm\text{interp}]$ is incoherent: absent features can be neither interpretable nor uninterpretable. The ϕ -Interpretability Parameter $[\pm\text{interp}]$ (contingent on $[\phi]$) asks: do ϕ -features on the pivot provide semantically contentful material to LF?

- $[-\text{interp}] \rightarrow$ **Type 2 ($[\text{u}\phi]$)**: ϕ -features lack semantic content at LF despite morphological presence (e.g., English residual agreement, or German/Icelandic/Faroese/Finnish's fully productive agreement)
- $[\text{+interp}] \rightarrow$ **Type 3 ($[\text{i}\phi]$)**: ϕ -features carry semantic content reaching LF (e.g., Italian, Hindi-Urdu, Swahili)

The three resulting types:

1. **Type 1 ($n\phi$)**: aspectual/evidential pivots, no person-number/gender inflection, discourse-based pro-drop, no expletives.
2. **Type 2 (ϕ -available + $[\text{u}\phi]$)**: valuation-based pivots, obligatory subjects and expletives, agreement richness independently variable (Type 2a: impoverished; Type 2b: rich but non-interpretive).
3. **Type 3 (ϕ -available + $[\text{i}\phi]$)**: matching-based Agr/Asp/Mood pivots, morphologically licensed pro-drop, rich agreement, no expletive.

The three-way typology follows *logically* from a binary-plus-dependent architecture, not from ad hoc tri-partitioning. Properties distinguishing ϕ -available from ϕ -less systems — morphology, case alternations, pivot-mediated subject licensing — should strongly correlate across languages; §3.5 tests this directly.

Diachronically and in acquisition, ϕ -availability is the more fundamental, earlier-acquired, and more change-resistant choice; ϕ -interpretability may shift within ϕ -available languages through erosion or reanalysis without loss of ϕ -based finiteness (exactly the Type 2b pattern; §3.8). Parametric dependency correspondingly reduces the acquisition search space: if ϕ -Availability = $[-\text{Available}]$, ϕ -Interpretability need not be evaluated at all. This predicts staged acquisition, with primary-parameter values converged on before secondary values become detectable.

3.4 Bimodal Agree: Formal Characterization

The two-level architecture determines how Agree proceeds, yielding fundamentally different licensing mechanisms for ϕ -available and ϕ -less systems.

In ϕ -available systems, the pivot bears $[\text{i}\phi]$ or $[\text{u}\phi]$ and establishes ϕ -dependencies with arguments bearing $[\text{i}\phi]$. Bimodal Agree involves matching of ϕ -features with valuation of $[\text{uCase}]/[\text{uEPP}]$; Unimodal Agree values $[\text{uCase}]/[\text{uEPP}]$ while ϕ -features are either valued or absent. Case licensing proceeds through the same operation as ϕ -checking; subject position is determined by the need to satisfy both ϕ -agreement and EPP, yielding specifier-movement in most systems or head-movement of the pivot itself, as in Warlpiri.

Compatibility between probe and goal requires overlapping ϕ -specifications.

Incompatibility arises from feature deficit (goal lacks ϕ entirely), feature clash (probe's specification finds no match), or intervention. In valuation mode, the probe's $[\text{u}\phi]$ are valued against the goal's $[\text{i}\phi]$ and deleted from the narrow syntax. In matching mode, the probe's $[\text{i}\phi]$ and the goal's $[\text{i}\phi]$ are co-indexed and both persist to the interfaces.

In $[n\phi]$ systems, Agree proceeds without ϕ -interaction at all: the pivot bears only $[\text{iCase}]$ and $[\text{uEPP}]$, values the argument's $[\text{uCase}]$ directly, and satisfies $[\text{uEPP}]$ through obligatory aspectual/evidential projection. The argument's ϕ -features remain unprobed, persisting to LF as freestanding features recoverable only through discourse. This Unimodal/Bimodal Agree bifurcation is structurally required, rather than being merely descriptive: the presence or absence of ϕ -features on the pivot determines

whether Agree can proceed through ϕ -mediation at all, yielding two qualitatively different derivations that nonetheless satisfy the same universal requirement that some functional head must anchor case and EPP.

3.5 Quantitative Validation of the Hierarchical Parametric Organization

3.5.1 Three-Sample Validation Strategy

To empirically test the prediction that the ϕ -state organizes morphosyntactic variance, the employed validation strategy consists of three-sample, assessing both internal consistency and generalizability of the predictions:

Sample 1: Core Languages (N=7) — The core sample consists of Balti, English, Hindi, Urdu, Italian, Swahili, and Warlpiri. These languages are intensively documented as primary data (except Warlpiri) consisting of native speakers judgements. The languages belonging to the core group are maximally diverse. They represent all the three predicted parametric types across six language families/subfamilies from five continents. Each of these core languages is tested with twelve syntactic probes.

Sample 2: Outgroup Languages (N=75) — For the maximal genealogical diversity and typological coverage, I expanded the outgroup sample to include seventy-five languages from twenty-one different families. All the generalizations are tested over this broader typological space including edge cases and mixed systems.

Sample 3: Combined Sample (N=82) — In order to further reinforce that the core and outgroup patterns reflect the same underlying dimension, we unified both the samples and ran the statistical tests. Combined sample consisting of core and outgroup languages validates that the results are not artifacts of the sample selection.

Methodological triangulation: each of three sample was analyzed quantitatively via (1) Principal Component Analysis, (2) polychoric correlation analysis, and (3) Multiple Correspondence Analysis. In addition to these statistical tests, I employed a fourth, independent layer of evidence obtained from corpus-derived validation via Universal Dependencies. This fourth layer of evidence is introduced in §3.5.5, providing a test of the theory's predictions using data the theory played no role in generating.

3.5.2 Primary Result: High Dimensional Concentration

The application of the four variables including *Phi_Availability*, *Rich_Agreement*, *Morph_ProDrop*, *No_Expletive* strongly support the prediction that a single dimension shall play a dominant role in the organization of the morphosyntactic variance:

- PC1 = **96.60%** (eigenvalue = 3.864)
- PC2 = **3.40%** (eigenvalue = 0.136)
- Cumulative = **100.00%**

The concentrated variance ratio of 96.60:3.40 strongly validates the MPH's prediction: a primary dimension (ϕ -state) dominates morphosyntactic organization and relegates secondary factors to negligible residual status.

Variable contributions to PC1: It is crucial to note that *Phi_Availability*, *Morph_ProDrop*, and *No_Expletive* contribute equally (25.61% each), confirming the deterministic bundling. However, *Rich_Agreement* contributes less than them (23.17%) which statistically reflects the theoretically important Type 2b exceptions, e.g., German, Icelandic, Faroese, Finnish having rich morphology but uninterpretable ϕ -features.

PC2 (3.40%) captures the residual: *Rich_Agreement* dominates it (76.83% contribution), distinguishing Type 2a (English, Dutch, French: degraded morphology) from Type 2b (German, Icelandic, Faroese, Finnish: robust paradigms) within the same [$u\phi$] category. The small magnitude confirms this within-type variation is subordinate to the primary parametric-type distinction.

3.5.3 Perfect and Near-Perfect Correlations

MPH predicts deterministic bundling for the core diagnostics. Under the reflective model (§3.1), these are convergent evidence that all four indicators track a shared latent φ -state, not pairwise causal relationships among the indicators themselves.

- **Phi_Availability $\leftrightarrow$ Morph_ProDrop: $r = 1.000$** (exact, 82/82 languages)
- **Phi_Availability $\leftrightarrow$ No_Expletive: $r = 1.000$** (exact, 82/82 languages)
- **Phi_Availability $\leftrightarrow$ Rich_Agreement: $r = 0.908$** (strong but, as predicted, imperfect — Type 2b exceptions)

Polychoric correlation analysis — appropriate for the underlying ordinal structure — returns all pairwise coefficients at or near 1.000, including Rich_Agreement. This reflects a genuine empirical fact (near-total categorical determinism in the corrected 82-language sample: essentially no counter-examples remain in any pairwise contingency table) but is reported here with an important statistical caveat: at this level of determinism, polychoric maximum-likelihood estimation is a boundary solution rather than an interior estimate, and should be interpreted as *"no counter-examples were observed across 82 languages for any pairwise combination,"* not as an independently informative correlation coefficient distinct from the Pearson values above.

3.5.4 Methodological Robustness

Cross-sample stability:

- Core languages (N=7): PC1 = 99.91%, PC2 = 0.09%
- Outgroup (N=75): PC1 = 96.34%, PC2 = 3.66%
- Combined (N=82): PC1 = 96.60%, PC2 = 3.40%

The 0.26-percentage-point gap between Outgroup-only and the full sample confirms the result is not an artifact of Core-language selection.

Leave-one-out jackknife analysis (82 iterations) confirms exceptional stability: range [96.5%, 97.4%] (0.9% spread), mean 96.6% (matching the full-sample estimate), SD = 0.176% — all 82 iterations successful. Maximum influence is shared across German, Icelandic, Faroese, and Finnish (the Type 2b cluster), consistent with their theoretically-motivated boundary status rather than reflecting any single leverage point.

Multiple Correspondence Analysis provides independent validation under an unordered-categorical treatment: Dimension 1 = 55.45%, Dimension 2 = 42.86%, cumulative = 98.31%. A quadratic regression of Dim 2 on Dim 1 ($R^2 = 0.966$) confirms Dim 2 is a Guttman/horseshoe artifact of representing near-deterministic ordinal categories on orthogonal axes, not independent structure — MCA thus corroborates the same single-dimension φ -state identified by PCA.

Study	Languages	Features	Method	PC1
Bickel & Nichols (2005)	72	~20	Factor analysis	45%
Dunn et al. (2011)	301	125	PCA	52%
Current study	82	4	PCA	96.60%

A **51.60**-percentage-point improvement over Bickel & Nichols and a **44.60**-point improvement over Dunn et al. reflects two principled differences: (1) **theory-driven targeting** — diagnostics were explicitly predicted to bundle from interface-level parameter specifications, whereas prior studies aggregated heterogeneous features without parametric predictions; (2) **interface-level parameter** — φ -Interpretability operates at the syntax-semantics interface with cascading morphological, syntactic, and interpretive consequences, producing tighter clustering than purely syntactic parameters with localized effects.

3.5.5 Independent Corpus Validation via Universal Dependencies

The internal statistical structure above (§3.5.2–3.5.4) is, by construction, computed from hand-coded values. To guard against the possibility that this structure reflects coding artifacts rather than genuine linguistic organization, Phi_Availability was independently tested against raw morphological data from the Universal Dependencies treebank corpus (v2.17; 339 treebanks, 186 languages) — a resource built by hundreds of annotators over two decades with no relationship to MPH.

A corpus-derived φ -index — a mechanical, zero-learned-parameter count of Person/Number/Gender feature co-occurrence on finite verbs — was computed for 80 of the 82 sample languages (Swahili and Balti lack UD treebanks and were validated instead via primary elicitation; §1.6.1). Classification thresholds were fixed a priori and never fit to the gold labels.

Results: φ -availability (0.1 vs. {0.2, 1.0}) is recovered at 97.5% accuracy (AUC=0.998); φ -interpretability (0.2 vs. 1.0) at 95.0% accuracy (AUC=0.966). Critically, all four residual misclassifications are independently explicable rather than unexplained noise, and fall into two theoretically distinct categories:

- **Norwegian and Swedish** (false negatives on availability): both have completely lost verbal person-marking, morphologically indistinguishable from genuinely phi-less languages — yet the same syntactic diagnostics that identify German, English, and Dutch as [u φ] (obligatory subjects, obligatory expletives) correctly place them at 0.2, not 0.1.
- **Icelandic, German, and Finnish** (false positives on interpretability): Icelandic, German and Finnish languages represent the languages which retain genuinely rich agreement (Rich_Agreement=1) despite the uninterpretability of φ -features. It establishes that the richness does not entail interpretability, the central empirical claim of Type 2b. This outcome is theoretically expected. φ -index quantifies the richness of overt agreement morphology, whereas φ -interpretability concerns the grammatical status of that morphology. Rich agreement is therefore necessary for φ -interpretability but is not sufficient to establish it.

A battery of morphology-only diagnostics cannot be expected to distinguish either of these two cases from Type 3, precisely because the pro-drop and expletive diagnostics provide access to a distinction that correctly classifies all five languages, regardless of whether their surface agreement is rich or entirely absent. This limitation is therefore not a shortfall of the corpus test; rather, it independently confirms that φ -interpretability performs genuine theoretical work beyond what morphological richness alone can predict.

This constitutes a second, independent line of evidence for the φ -state, derived from data that played no role in formulating the theory and evaluated against a threshold fixed before the test was conducted. It therefore provides precisely the kind of independently testable prediction required by a Popperian criterion of falsifiability (§1.9.2).

3.5.6 The Organization of the Locus of the EPP

The results point to a clear hierarchical structure rather than a set of independent parameters. A language's φ -state is the underlying parameter, and once that state is fixed, the four observed reflexes—Phi_Availability, Rich_Agreement, Morph_ProDrop, and No_Expletive—pattern together as its consequences. They should not be understood as successive links in a causal chain. The strong clustering of these variables on PC1, together with the much smaller residual captured by PC2, is exactly what this architecture predicts.

The 96.60:3.40 split between PC1 and PC2 is hard to square with the idea that φ -availability and φ -interpretability vary independently. If they were genuinely free-standing dimensions, it would have been natural to expect each to leave a substantial imprint on the data. Instead, the overwhelming share of the variance—96.60%—falls on PC1, while PC2 accounts for just 3.40% dominated by Rich_Agreement. A distribution closer to 50:50 would be much more in line with independent parameters. The observed pattern points in the opposite direction: the two dimensions are not contributing on equal terms, and the structure captured by PC1 clearly dominates the typological variation. This concentration is what we would expect if φ -availability establishes the initial structural division and interpretability is evaluated only after that division has been made. In other words, a language first must be φ -available before the question of whether its φ -features are interpretable can arise. Availability thus sets the domain of variation for interpretability rather than determining its value.

The modest contribution of PC2 follows from the same structure. Type 1 languages are uniformly [–available], so there is no further interpretability contrast to recover within that group. Most of the residual variation consequently comes from the φ -available languages. It includes the Type 2a/2b morphological contrast, the part of the Type 2–Type 3 distinction left unexplained by PC1, and gradience in morphological richness within the individual types. PC2 therefore captures distinctions that are real but structurally subordinate. They emerge inside the domain already delimited by φ -availability, rather than competing with it as an independent source of typological organization.

3.5.7 Three Discrete Typological Clusters

The distribution of the languages is strikingly discrete. Rather than forming a continuous morphological continuum, they group into three clusters corresponding to the three MPH types.

Type 1 [– φ] (N=17, 20.7%) comprises Chinese, Cantonese, Japanese, Korean, Thai, Vietnamese, Balti, Tagalog, Cebuano, Yoruba, Indonesian, Javanese, Khoekhoe, Coptic, Bambara, Haitian Creole, Malayalam, and others. What unites these languages is the absence of φ -agreement together with discourse-licensed null arguments and no requirement for an overt expletive.

Type 2 [$u\varphi$] (N=12, 14.6%) is internally divided along morphological lines. Type 2a, represented by English, Dutch, French, Danish, Norwegian, Swedish, and others, has relatively poor agreement morphology, whereas Type 2b, including German, Icelandic, Faroese, and Finnish, may exhibit considerably richer agreement. This difference in surface morphology, however, does not translate into a difference in φ -status. The two groups pattern together on the diagnostics that are central to the present analysis: neither supports ordinary pro-drop, and both require an overt expletive in the relevant constructions. In other words, richer agreement in Type 2b does not by itself produce the grammatical properties associated with interpretable φ -features.

Type 3 [$i\varphi$] (N=53, 64.6%) presents a markedly different picture. The group includes Italian, Spanish, Russian, Hindi, Urdu, Swahili, Greek, Polish, Arabic, Basque, Georgian, and many others. Across these languages, the same cluster of properties recurs: φ -features are interpretable, agreement morphology is rich, null subjects are freely available, and there is no comparable requirement for an overt expletive. The important point is that their richer morphology is not, by itself, what sets them apart from Type 2 languages. It goes hand in hand with a different φ -state, and it is this difference that accounts for the syntactic properties predicted by the MPH.

The effect-size results make the discreteness of these groupings even clearer. Type 1 and Type 2 are separated by a large effect ($d=2.18$), while the Type 2–Type 3 contrast is extraordinarily large ($d=16.96$). The Type 1–Type 3 comparison requires a different interpretation because Cohen’s d is undefined: both groups have zero within-type variance on PC1. That outcome is actually more informative than a merely large finite effect. The two groups are internally homogeneous on this dimension and occupy non-overlapping positions, so there is no within-group dispersion against which a conventional standardized effect can be calculated.

3.6 Pro-Drop: Two Licensing Mechanisms

The typology also changes the way pro-drop itself should be understood. The MPH does not treat null subjects as the product of a single licensing mechanism operating uniformly across languages. The source of the licensing depends on the φ -state. φ -availability establishes the relevant domain, and φ -interpretability then determines how subject recoverability is achieved within that domain. This distinction becomes important because superficially identical cases of pro-drop need not have the same grammatical basis.

Pivot-mediated pro-drop occurs in $[i\varphi]$ systems such as Swahili, Italian, Hindi-Urdu, and Warlpiri. In these languages, interpretable φ -features are realized at PF through agreement morphology, allowing the reference of an unexpressed subject to be recovered without an overt DP. The crucial point is that this recoverability does not depend on discourse continuity. Null subjects can occur even when they introduce a new discourse referent, and the licensing system does not require topic continuity or depend on person-hierarchy effects. The relevant information is supplied by the agreement system itself: because φ -features are morphologically realized and interpretable, the missing subject can be identified through the morphology rather than inferred primarily from pragmatic context.

Discourse-mediated pro-drop occurs in $[n\varphi]$ systems (Balti): the pivot lacks φ -features entirely, so φ -information is never probed and remains recoverable only pragmatically. Balti *thong-yod* ('is seeing') specifies no person or number; the null subject is licensed by discourse salience alone. This mechanism shows the opposite signature: strong discourse sensitivity, a person hierarchy favouring egophoric contexts, and resistance to introducing new referents via null subjects.

$[u\varphi]$ systems permit neither mechanism. In English, T's uninterpretable φ must be valued by an overt goal and is deleted post-valuation — blocking pivot-mediated pro-drop (no recoverable morphology survives to PF) and discourse-mediated pro-drop alike ($u\varphi$ still forces syntactic probing, requiring an overt subject or expletive). This blocking holds uniformly across $[u\varphi]$ systems regardless of agreement richness: German (Type 2b) blocks pro-drop despite rich morphology exactly as English (Type 2a) does with impoverished morphology, since the blocking factor is interpretability, not paradigm size — the clearest possible illustration that Rich_Agreement dissociates from the theory's other three diagnostics.

The two mechanisms, as well as their absence in $[u\varphi]$ systems, are not added as independent assumptions; they follow from the architecture itself. This makes the analysis empirically useful, since it yields concrete predictions about discourse sensitivity and person-hierarchy effects while also predicting that pro-drop should not simply track the richness of overt agreement.

3.7 Integration with Phase Theory

Phase theory provides the natural ground to accommodate the pivot and its associated hierarchical parametric organization. Under the Phase Impenetrability Condition

(Chomsky 2000, 2001), the pivot takes over the functions that are traditionally associated with T.

It is the first phase-external head to probe the ν P edge, while its featural specification is determined by the interaction of the ϕ -Availability and ϕ -Interpretability Parameters. In ϕ -available [u ϕ] systems, the phase-external pivot initiates downward search once ν P transfers. English T, bearing [u ϕ], [iCase], [uEPP], probes the transferred ν P edge; [u ϕ] is valued by the goal's [i ϕ] and deleted; [iCase] values [uCase] as nominative; [uEPP] attracts the DP to Spec-TP, or triggers expletive insertion when no thematic subject is available, since [u ϕ] still requires a bearer even when semantically vacuous.

In ϕ -available [i ϕ] systems, the pivot establishes matching relations at the ν P edge instead. Swahili Agr, bearing [i ϕ], [iCase], [uEPP], matches its interpretable ϕ with the subject DP's through co-indexation; both persist to the interfaces; [iCase] values [uCase]; [uEPP] is satisfied through the matching relation itself, triggering overt movement only as information structure requires. At PF, the interpretable ϕ -features on the pivots surface as subject prefixes, licensing null subjects.

Warlpiri instantiates a variant where the pivot originates *within* the ν P phase and establishes matching relations with both subject and object before head-moving to second position — satisfying the EPP without phrasal movement to a specifier, and explaining the coexistence of free DP order with a fixed auxiliary position. The moved auxiliary's interpretable ϕ licenses pro-drop for both arguments independently.

In ϕ -less systems, the phase-external pivot probes for [uEPP] only. Balti Asp/Evid, bearing no ϕ but carrying [iCase] and [uEPP], values the subject's [uCase] directly as ergative or nominative depending on the pivot's own aspectual properties (perfective $\rightarrow$ ergative; non-perfective $\rightarrow$ nominative). No ϕ -probing occurs; [uEPP] is satisfied through obligatory aspectual/evidential projection; the subject's interpretable ϕ -features remain unprobed, available only for discourse-based interpretation.

Phase theory thus supplies the computational framework within which ϕ -availability and ϕ -interpretability determine specific licensing mechanisms: the phase edge fixes where probing initiates, the pivot's featural specification fixes what feature-checking is possible, and interpretability fixes whether checking proceeds through valuation-with-deletion or matching-with-persistence.

3.8 Empirical Consequences and Typological Predictions

The two-level architecture generates specific, testable predictions about possible and impossible language types, property clustering, and diachronic trajectories — now tested against two independent evidence sources rather than one.

The framework predicts that certain property combinations should be unattested.

Languages should not show morphologically-licensed pro-drop while also requiring obligatory expletive insertion, since pivot-mediated pro-drop and expletive-obligatoriness diagnose interpretable and uninterpretable ϕ -features respectively, and the two cannot co-occur on the same pivot. This prediction holds without exception across the full 82-language sample — Phi_Availability, Morph_ProDrop, and No_Expletive are empirically inseparable at $r=1.000$ (§3.5.3) — and is independently corroborated by the UD corpus test (§3.5.5). Rich_Agreement is deliberately excluded from this inviolable core, since agreement morphology can survive independently of interpretability through diachronic retention (§3.5.6, §5.5): rich person-number morphology may therefore co-occur with expletive-obligatoriness and blocked pro-drop, as it does in German, Icelandic, Faroese, and Finnish. Rather than weakening falsifiability, this states it more precisely. The framework predicts deterministic bundling for three of the four diagnostics and a principled, independently motivated

dissociation for the fourth, and both halves of that prediction hold up, on both statistical and corpus evidence. Languages should also not lack person-number morphology entirely while showing pivot-mediated pro-drop, nor show discourse-insensitive pro-drop without any agreement morphology. These predictions create falsifiable boundaries on cross-linguistic variation.

The framework predicts tight within-type clustering, evident in PC1's 96.60% concentration. φ -unavailable systems should uniformly lack person-number paradigms, show aspectual/evidential clause-defining morphology, exhibit aspect-conditioned (not φ -conditioned) case alternations, and permit discourse-mediated pro-drop with person-hierarchy effects. φ -available [$i\varphi$] systems should show rich agreement, unrestricted pro-drop, no expletive requirement, and φ -interacting case systems. φ -available [$u\varphi$] systems should require overt subjects or expletives, block pro-drop, and tie case licensing to φ -valuation — but agreement richness within this type is **not** uniform: Type 2a (English, Dutch, French) shows impoverished agreement; Type 2b (German, Icelandic, Faroese, Finnish) retains rich agreement despite uninterpretable status, confirming that agreement richness alone is not a reliable interpretability diagnostic. Clustering should be tighter within types than across them, and tighter on the three deterministic diagnostics than on Rich_Agreement specifically.

The framework also makes predictions about how these properties should emerge during acquisition. If a child is acquiring an [$i\varphi$] language, agreement should become productive relatively early, and null subjects should be available from the early multiword stages rather than appearing only after a prolonged developmental period. On this view, there should also be no intermediate stage in which children systematically omit expletives that the adult grammar requires. Deen's (2003) longitudinal study of Swahili is consistent with this prediction: children show adult-like agreement and productive pro-drop from the beginning of multi-clausal production, with no identifiable expletive stage. Comparable early development has been reported for Italian and Spanish as well (Hyams 1986; Rizzi 1994; Guasti 2002).

The developmental pattern should be different in [$u\varphi$] languages. Here, the relevant factor is not how rich the adult agreement paradigm happens to be, but the fact that φ -features are uninterpretable. The prediction, therefore, is that Type 2a and Type 2b languages should show broadly similar acquisition patterns despite their very different surface morphology. Children should pass through an Optional Infinitive stage (Wexler 1998), followed by a period in which overt subjects are not yet consistently required, together with a relatively extended stage of expletive omission. English provides the familiar case, but German is particularly revealing. Although German has a rich agreement paradigm in the adult grammar, German-speaking children nevertheless show an Optional Infinitive stage comparable to that found in English. The developmental evidence therefore points away from morphological richness as the decisive factor and instead supports the claim that acquisition is sensitive to the underlying interpretability of φ -features.

Children acquiring [$n\varphi$] systems should master aspectual/evidential markers early, produce discourse-based pro-drop from early stages, and may temporarily overgeneralize null subjects to discourse-inappropriate contexts before pragmatic licensing stabilizes (Deen 2003: §6).

The framework predicts asymmetric diachronic change. Shifts in φ -interpretability within φ -available systems (paradigm erosion or enrichment) should not require restructuring clause architecture; shifts between φ -available and φ -unavailable settings require more radical reanalysis. Surface changes in agreement richness should therefore

occur more frequently than changes in fundamental clause-anchoring mechanism — a prediction directly confirmed by Type 2b's existence: German, Icelandic, Faroese, and Finnish retained ancestral agreement morphology after their pivots had already become uninterpretable, exactly the asymmetric-change pattern predicted here, independently visible in both the statistical clustering (§3.5.2) and the corpus validation (§3.5.5).

4. Comparison with the Existing Models

4.1 Upward Agree: Bjorkman & Zeijlstra 2019

Upward Agree proposes that agreement operations proceed via upward search (Bjorkman & Zeijlstra 2019, p. 488–492): goals bearing interpretable features [iF] probe upward to value uninterpretable features [uF] on c-commanding functional heads (Zeijlstra 2012, p. 497–500), reversing the directionality of Chomsky's (2000, 2001) downward Agree (Bjorkman & Zeijlstra 2019, p. 359–360). This eliminates stipulations about search direction and captures long-distance agreement phenomena where lower elements appear sensitive to higher functional specifications.

The fundamental challenge: Upward Agree maintains that φ -agreement universally involves [u φ] on functional heads being valued by [i φ] on DPs through upward search. The MPH, however, shows systems where pivots bear i φ (not u φ) and establish matching relations, or are φ -less with no φ -probing whatsoever. These patterns pose challenges to Upward Agree's universal on-heads architecture.

Empirical Challenge 1: Balti φ -less Pivots

Balti aspectual-evidential pivots obligatorily project and license case without any φ -features:

- (1) *Go-ain yod*
 go-EGO EVID.NPST
 '(I) am going (I know directly) '

The aspectual-evidential auxiliary *yod* obligatorily projects and values ergative case on subjects, yet shows no person, number, or gender distinctions. The form remains invariant across all person-number-gender combinations.

The challenge for Upward Agree: Under Bjorkman & Zeijlstra's framework, when a functional head needs to license subjects (assign case, satisfy EPP), it must bear [u φ] that gets valued by the subject's [i φ] through upward search. Balti Asp should bear [u φ] and receive valuation from subject DPs. The framework allows for default insertion when upward probing fails (Bjorkman & Zeijlstra, p. 372–374), predicting that Balti should show consistent default φ -morphology (e.g., third person singular, cross-linguistically common as default (Bjorkman & Zeijlstra, p. 369–371)).

What the Balti data show instead: there is no evidence of φ -morphology on the pivot—not even a default form that could be interpreted as failed agreement. Person-number marking is simply absent. The auxiliary *yod* is invariant because it expresses aspect and evidentiality (perfective + egophoric), not because it carries a default third-person specification.

Balti also allows extensive pro-drop in the absence of any φ -agreement morphology:

- (2) *(Pro) aasho thong-sad pa*
 woman see-PST.PFV
 '(Someone) saw the woman'

This pattern is difficult for Upward Agree to accommodate. Under that approach, pro-drop is expected to follow from successful upward valuation (Bjorkman & Zeijlstra 2019: 367–369), with overt φ -morphology providing the morphological reflex needed to license a null subject (Zeijlstra 2012: 502–503). Balti presents the opposite configuration: null subjects are possible even though the pivot carries no φ -morphology

at all. What the grammar provides instead is aspectual-evidential marking, while the identity of the null argument can be recovered from discourse. Pro-drop is therefore not tied to φ -agreement in Balti. This makes the Balti pattern particularly important: null-subject licensing is independent of φ -valuation, contrary to the prediction that $[\text{u}\varphi]$ valuation must be involved in argument licensing.

The MPH analysis: On the MPH account, Balti Asp does not carry φ -features at all. The pivot values case through $[\text{iCase}]$ and satisfies the EPP through $[\text{uEPP}]$, with the subject DP moving obligatorily to Spec-Asp/EvidP. Because the pivot has no φ -features, there is no φ -probing and consequently no φ -agreement relation to establish. This allows the invariant auxiliary to be analyzed for what it actually expresses, rather than being treated as a default agreement form.

The subject DP, meanwhile, retains its $[\text{i}\varphi]$ features as independently interpretable features available at LF. A null subject can therefore be licensed when the discourse provides enough information to recover its reference; morphological φ -agreement is not needed for that purpose. The analysis also avoids a problem that would otherwise arise under a valuation-based account: there is no failed valuation and hence no derivational crash, nor is there any need to posit default morphology. There are simply no φ -features on the pivot that require valuation or default realization in the first place.

Empirical Challenge 2: Hindi-Urdu $[\text{i}\varphi]$ Pivots

Hindi-Urdu Asp bears $[\text{i}\varphi]$ rather than $[\text{u}\varphi]$, engaging in matching-based Agree rather than valuation:

(3) Imperfective:

(pro) roṭī khā-tā hai
bread.F eat-IMPF.M.SG be.3SG
'(He) eats bread'

The imperfective suffix *-tā* shows masculine singular morphology reflecting Asp $[\text{i}\varphi]$ matching with nominative subject's $[\text{i}\varphi]$. This licenses morphologically mediated pro-drop.

The challenge for Upward Agree: Under Bjorkman & Zeijlstra (2019, p. 360–361), functional heads bear $[\text{u}\varphi]$ that get valued by DP's $[\text{i}\varphi]$. Hindi-Urdu reverses this: Asp bears $[\text{i}\varphi]$ (evidenced by rich paradigmatic morphology: *-tā* M.SG, *-tī* F.SG, *-te* M.PL) and matches with DP's $[\text{i}\varphi]$. There is no $[\text{u}\varphi]$ on Asp requiring upward valuation. Instead, both Asp and DP bear interpretable features that must be compatible and co-indexed through matching.

Upward Agree could respond that DPs bear $[\text{u}\varphi]$ that probe upward to value against Asp $[\text{i}\varphi]$, but this is unmotivated: DPs inherently bear interpretable φ -features (person, number, gender) that do not require valuation. The Hindi-Urdu pattern is more naturally captured by a bidirectional compatibility relation between Asp's $[\text{i}\varphi]$ and the DP's $[\text{i}\varphi]$ than by a strictly upward valuation mechanism. The relevant features must be checked against one another, rather than simply being valued in one direction.

The asymmetry introduced by the perfective makes the problem even clearer:

(4) Perfective:

(pro) roṭī khā-ī
bread.F.SG eat-PFV.F.SG
'(Someone) ate bread'

Here, perfective Asp continues to carry $[\text{i}\varphi]$. Yet the subject is marked ergative, which prevents it from participating in the relevant matching relation. Asp instead matches with the nominative object *roṭī*, producing the feminine singular form *khā-ī*. For Upward

Agree to derive this result, ergative case would have to be treated as an intervention effect that stops the upward search from reaching the subject. The analysis would then need some additional mechanism allowing the DP to search laterally or downward toward the object. At that point, however, the central assumption of a uniformly upward Agree relation is no longer maintained.

Upward Agree remains useful for cases in which a lower element is demonstrably sensitive to a higher specification, as in negative concord and sequence-of-tense phenomena. The difficulty arises with systems in which the pivot itself has no ϕ -features but can nevertheless license arguments. Such cases suggest that argument licensing cannot in general be reduced to ϕ -mediated upward valuation.

4.2 Flexible Agree: Preminger 2014

Flexible Agree takes a different approach to failed ϕ -valuation. For Preminger (2014: 6–9), failure to value a probe's ϕ -features does not necessarily cause the derivation to crash. If a $[u\phi]$ probe does not find an appropriate goal, the derivation may proceed, with the unvalued features either eliminated at the interfaces or expressed through default morphology (Preminger 2014: 86–92, 126–133). This provides a way of handling languages in which agreement is optional or in which a default form appears even when a potential agreement controller is structurally available.

Balti's feature composition poses problem for Flexible Agree as well. The problem for Flexible Agree goes beyond a question of whether agreement succeeds or fails in the case of a specific language. Flexible Agree is founded on the assumption that the morphosyntactic pivot always carries $[u\phi]$. This assumption leads to the conclusion that differences across languages can be derived from whether those features are successfully valued. However, the Balti facts pose serious challenge for this fundamental assumption because the morphosyntactic pivot has no ϕ -features to value at all, yet its morphology is neither absent nor optional. Aspectual and evidential distinctions are expressed obligatorily and can produce a morphologically rich paradigm without involving ϕ -agreement in any form. Balti therefore presents a case in which rich verbal morphology and pro-drop occur independently of ϕ -valuation. The latter is licensed through discourse recoverability, not through agreement morphology, making it difficult to treat the pattern simply as a consequence of failed valuation.

Empirical Challenge 1: Balti Obligatory Evidentials Without ϕ -Features

Balti displays a system that makes this distinction between discourse-based and pivot-mediated pro drop particularly clear. Its aspectual-evidential system is obligatory and morphologically well developed, but none of its forms carries person-number agreement:

(5) Balti evidential paradigm (all forms lack person-number distinctions)

- a. *go song-set* — '(He/she/they) went' — perfective direct
- b. *go-en yod* — '(He/she/they) is going' — progressive visible
- c. *go-ain yod* — '(I/we) am/are going' — progressive egophoric
- d. *go-chatkhikha song-fin* — '(He) must have gone' — inferential
- e. *go-song-fin* — '(They say) (he) went' — reportative

The egophoric form in (5c) is especially useful for separating evidential meaning from agreement. Although it can receive a first-person interpretation, that interpretation does not come from a first-person morphological marker. The form indicates the speaker's direct involvement or experience, from which the first-person reading follows naturally. Person is not morphologically encoded. The same holds across the paradigm: person, number, and gender remain invariant, while the overt contrasts express aspectual and evidential distinctions.

This is where the predictions of Flexible Agree become difficult to maintain. If the pivot carries [u ϕ] but fails to find a suitable goal, the theory permits either a default form to emerge—typically something like 3SG—or the absence of morphology when valuation does not succeed (Preminger 2014: 88–92). Neither description fits Balti. Its morphology does not look like agreement that has failed; nor does it disappear when ϕ -valuation is unavailable.

The contrast becomes particularly clear when the evidential system is considered as a whole. The relevant morphology is **obligatory**: finite clauses require evidential specification, and a bare *go* meaning ‘went’, without the appropriate evidential marking, is ungrammatical. The paradigm is also **rich**, distinguishing five evidential categories alongside aspectual contrasts. Yet throughout these distinctions there is no person-number morphology whatsoever. Nothing in the paradigm can reasonably be identified as a default agreement form. The morphology is doing something else: it is encoding aspect and evidentiality on a pivot that is entirely ϕ -less.

This creates a three-way mismatch with the expectations of Flexible Agree. The richness of the Balti paradigm cannot be the result of successful [u ϕ] valuation, because there are no ϕ -features on the pivot to value. Nor can it be reduced to default morphology, since the forms do not instantiate a default person-number value. What the paradigm actually spells out are aspectual and evidential properties of the pivot.

Pro-drop makes the contrast sharper still. Flexible Agree associates morphologically licensed pro-drop with successful ϕ -valuation (Preminger 2014: 251–258), since default morphology resulting from failed valuation would normally provide little information for identifying a null subject (p. 133). Balti nevertheless permits pro-drop in a system with no ϕ -agreement whatsoever. The recovery of the null argument therefore has to come from elsewhere—most naturally from discourse salience, topic continuity, or contextual recoverability. This is not simply a case of agreement failing to appear; it is a system in which ϕ -agreement is not part of the licensing mechanism in the first place.

Balti permits extensive pro-drop:

- (6) (*Pro*) *zan za-en yod*
 food eat-PROG EVID.NPST
 ‘(He/she) is eating food’

Pro-drop requires evidential + discourse context, rather than ϕ -agreement.

Flexible Agree cannot straightforwardly derive this: obligatory evidential paradigms without ϕ -interaction licensing discourse-based pro-drop suggests no ϕ -probing occurs, not failed valuation with defaults.

Empirical Challenge 2: Expletive Requirements

Yet another problem for the Flexible Agree is to explain why English requires expletives when no thematic subject is available, while Balti permits null subjects freely:

- (7) English vs. Balti weather predicates
 a. English: *(It) is raining. → Expletive obligatory
 b. Balti:
 Charfa ongain yoda? → No Expletive required
 Rain come PROG. DIR. Q
 ‘Is rain coming?’

In terms of Flexible Agree, English weather verbs lack thematic subjects, Balti employs the weather itself as the subject DP along with a common verb such as *ongain* (coming). For this purpose, the language-specific conditions determining whether failed valuation requires expletive insertion (English) or permits weather to serve as the subject itself

(Balti) must be stipulated by the framework. This loses the generalization that expletive requirements correlate with probe ϕ -specifications.

This contrast between the different behavior of English and Balti is derivable from the contrast between the featural architecture of English T and Balti Asp/Evid.

English T bears [$u\phi$] and, therefore, creates a valuation dependency requiring either a DP goal with [$i\phi$] or expletive insertion. Uninterpretable features must be eliminated before LF (Full Interpretation), forcing overt realization even with non-thematic expletives.

As Balti Asp bears [$n\phi$], it simply eliminates the valuation dependency entirely. EPP is satisfied by obligatory projection of aspectual-evidential morphology hosting the subject with no ϕ -features requiring checking. Subject ϕ -content is recovered from discourse when contextually salient. No expletive is needed because no ϕ -valuation is demanded.

Why Flexible Agree Cannot Simply Posit ϕ -unavailable Systems: It is possible for Flexible Agree to accommodate Balti by allowing pivots to optionally lack [$u\phi$] entirely, eliminating all valuation dependencies. However, this concedes the MPH's core claim: pivots vary in ϕ -specification [$n\phi$], [$u\phi$], [$i\phi$]) rather than universally bearing [$u\phi$] with parametric variation in valuation outcomes. Moreover, Flexible Agree's theoretical motivation—explaining why failed valuation doesn't crash—becomes vacuous if pivots can simply lack features that trigger valuation. The framework's explanatory force is universally dependent on universal [$u\phi$] on pivots; these pivots attempt valuation, with cross-linguistic variation arising from success/failure patterns rather than from featural specifications themselves.

Conclusion: As long as the pivots carry ϕ -active systems, Flexible Agree successfully handles variation within where valuation may succeed or fail. However, it faces challenge in dealing with languages having [$n\phi$] systems where pivots obligatorily project rich paradigms (evidentials) without any ϕ -interaction, license arguments through case alone, and permit discourse-based pro-drop without morphological ϕ -content. The Balti pattern requires recognizing that some pivots lack ϕ -features entirely, not that they bear [$u\phi$] with consistently failed valuation.

4.3 Feature Inheritance: Chomsky 2007, 2008; Richards 2007; Miyagawa 2010

Feature Inheritance proposes that ϕ -features originate on phase heads (C, v), and transfer to non-phase heads (T, V) where they trigger agreement (Chomsky 2007, p. 10–12, Chomsky, 2008, p. 138–143). Richards (2007, p. 138–142) extends this to successive-cyclic movement, while Miyagawa (2010, p. 59–66) allows ϕ -features to remain on C in discourse-configurational languages. The correlations between phasehood and agreement while maintaining T's central role in finite clauses is elegantly derived by the framework.

The fundamental incompatibility: The Feature Inheritance requires an obligatory TP projection in all finite clauses because finiteness correlates with ϕ -feature activity inherited to T (Chomsky 2008, p. 143–147). However, the MPH framework contradicts this assumption about the universality of T as the host of EPP. It demonstrates that finite clauses can, indeed, lack TP entirely when non-T heads serve as pivots.

Empirical challenge 1: Hindi-Urdu subjunctives (AspP without TP)

(8) *main chāhtā hūn [ki Abdullah rotī khā-e]*

I want.M.SG be.1SG that Abdullah bread eat-SBJV.M

'I want that Abdullah eat bread'

The embedded clause in (8) is finite because it licenses nominative *Abdullah*, forms independent conjuncts and is selected by finite complementizer *ki*, yet it contains only aspectual morphology and no TP. The subjunctive aspect marker *-e* contrasts paradigmatically with perfective *-ā* and imperfective *-tā*, instead of tense because the temporal interpretation of (8) depends solely on matrix *chāhtā hūn*, with tense expressed by optional auxiliaries (*thā* past, *gā* future). In contrast to non-obligatory tense, aspectual morphology is obligatory in Hindi-Urdu finite clauses. This distribution indicates Asp, not T, anchors the clause.

To maintain the uniformity in its predication, Feature Inheritance must claim the subjunctive aspect marker *-e* specifies T. However, such a position is in contradiction with the observed patterns. Firstly, the data clearly indicate that *-e* patterns aspectually, not temporally. Secondly, the embedded temporal interpretation⁴ is non-local, derived from the matrix verb rather than independent embedded T. And thirdly, cross-linguistically, subjunctive is aspectual in Hindi-Urdu but fused with T in Italian (*vada* SBJV.PRS vs. *andassi* SBJV.IMPF), indicating subjunctive does not universally occupy T.

Empirical Challenge 2: Italian Imperatives — MoodP Without TP

An observed fact in Italian is the occurrence of Mood independently of T. It rather provides a particularly clear case in which clause structure is organized around Mood with no T at all:

(9) *Vieni e siediti!*

come.IMP.2SG and sit.IMP.2SG=REFL

‘Come and sit down!’

As evident from (9), the imperative has its own mood morphology and behaves as an independent finite clause. However, it does not participate in the tense paradigm in the way an ordinary indicative clause does. In Italian, there is no imperative counterpart of past or future tense: **venne* cannot mean ‘past imperative’, nor can **verrai* mean ‘future imperative’. It is important to note that the syntax of the clause changes with mood. This change is evident from the Clitic placement which makes this especially visible:

(10)

a. *Lo vedo.*

it.CL see.IND.1SG

‘I see it.’

Indicative: proclitic

b. *Vedilo!*

see.IMP.2SG=it.CL

‘See it!’

Imperative: enclitic

The morphological alternation in (10a) and (10b) is not a minor difference. It shows the mood systematically changes with the position of the clitic. The clitic precedes the verb in the indicative but follows it in the imperative. Had T universally carried the relevant inherited ϕ -features, as proposed in the Feature Inheritance approach (Chomsky 2008: 143–147), why changing the mood of the clause should produce such a structural reorganization cannot be accounted for with the bounds of this theory. The Italian facts instead point to Mood as the functional head responsible for the relevant organization.

⁴ However, it must be noted that there can be clauses in which neither the matrix nor the embedded clauses contain T, i.e., *main-ne chāha [ki Abdullah roṭī khā-e]*. Neither of the clauses contains a temporal marker but both contain aspectual inflections. This further establishes the subservient nature of TP and superiority of Asp in Hindi-Urdu.

Positing imperative as a special instance of T may be possible response. But that move comes at a cost. It does not account for why imperatives lack the ordinary tense distinctions found elsewhere in the verbal system, nor does it capture the broader range of syntactic effects associated specifically with mood. The same mood-sensitive organization is visible in subject position, negative imperatives, complementizer selection, and clitic placement (see Appendix E.4.1). Treating imperative Mood as simply another realization of T would therefore restate the problem rather than explain why these properties consistently follow Mood.

Miyagawa (2010) does not remove the problem

Miyagawa's (2010) extends the framework and allows ϕ -features to remain on C in discourse-configurational languages. However, despite this extension, the broader architecture still ties ϕ -features to C or T as the central locus of clause structure (2010: 32–41). The cross-linguistic evidence considered here points to a less uniform distribution. In Hindi-Urdu, ϕ -features are associated with Asp, as shown by the gender-number contrasts that accompany aspectual morphology: *khā-yā* 'ate' M.SG versus *khā-ī* 'ate' F.SG. Italian provides a different configuration, in which agreement is tied to Mood, including the agreement patterns found in the subjunctive system. Balti goes even further still and allows its aspectual-evidential pivots carry no morphological ϕ -features at all.

To derive these patterns within Feature Inheritance, one would have to allow the relevant features to be inherited from C to Asp in Hindi-Urdu, from C to Mood in Italian, and effectively to disappear as ϕ -features altogether in Balti. At that point, however, the analysis has moved away from the Feature Inheritance. The empirical generalization supported by the data contradicts a uniform C/T-centered architecture. Instead, it proposes that different functional heads can serve as the structural pivot of the clause in different languages.

This is the broader difficulty for Feature Inheritance. If T universally provides the locus for finiteness-related features, whether those features are inherited from C or are directly associated with T, the theory naturally predicts a TP-centered structure for finite clauses. The MPH makes an even stronger claim that the licensing functions normally attributed to T can instead be carried by different functional heads, called morphosyntactic pivots, across different languages. Italian imperatives provide a concrete illustration of this possibility. The issue is, therefore, not whether T exists as an abstract category, but whether T serves as the universal structural anchor of finiteness and agreement. The evidence presented here suggests that it does not.

4.4 Multiple Agree (Hiraiwa: 2001, 2005)

Instead of single probe with agreement relation with single goal, Multiple Agree allows a single probe to establish agreement relations with multiple goals simultaneously. It has been very successful in capturing polypersonal agreement patterns where morphology reflects both subject and object ϕ -features (Hiraiwa 2001, p. 68–70; 2005, 99–108; Nevins 2011).

The core challenge: The case-conditioned selective matching with a single goal in Hindi-Urdu poses a serious problem for the Multiple Agree. Hindi-Urdu demonstrates that apparent "object agreement" involves case-conditioned selective matching with one accessible goal, not simultaneous Multiple Agree with all arguments.

Hindi-Urdu: case-conditioned selective matching

Hindi-Urdu Asp carries [i ϕ] in both imperfective and perfective clauses, but the arguments available for ϕ -matching are not the same in the two aspectual environments. Case is what creates the crucial asymmetry.

In the imperfective, the nominative subject can match with interpretable ϕ -features of Asp:

(11) *Abdullah roṭī khā-tā hai*

Abdullah bread.F.SG eat-IMPF.M.SG be.PRS

'Abdullah eats bread' → *-tā* matches masculine singular *Abdullah*

The verbal inflection '*tā*' matches the $i\phi$ on the subject. However, the situation changes in the **perfective**. Ergative marker '*ne*' on the subject prevents it from entering the relevant ϕ -matching relation with Asp (Bobaljik 1993; Laka 1993; Woolford 1997). Due to this blockage of agreement, the object, when accessible, becomes the argument that determines the form of the aspectual morphology:

(12) *Abdullah-ne roṭī khā-ī*

Abdullah-ERG bread.F.SG eat-PFV.F.SG

'Abdullah ate bread' → *-ī* matches feminine *roṭī*, not ergative *Abdullah* M.SG

In (12), it is the object which agrees with the aspectual inflection. Two further patterns make the nature of this relation clearer.

Critical pattern 1: Default morphology when no accessible object is available

Consider a clause in which the object is non-specific or simply not expressed:

(13) *Abdullah-ne (kuch) khā-yā*

Abdullah-ERG something eat-PFV.M.SG

'Abdullah ate (something)' → default masculine (failed matching)

The masculine singular form here is not evidence that Asp has successfully matched with the ergative subject. Rather, it appears when no accessible argument supplies the relevant ϕ -values. This is difficult to reconcile with a Multiple Agree analysis in which Asp probes the subject and object simultaneously. If both arguments were being probed, we would expect some morphological indication of those multiple relations, even when one of the goals ultimately proved inaccessible (Hiraiwa 2005: 118–135). What we actually find is simpler: when matching fails, a uniform default form surfaces.

Critical pattern 2: Only one instead of multiple arguments determines the morphology

When the ergativity blocks the subject agreement and the object is accessible, its ϕ -features determine the morphology, with no morphological contribution from the ergative subject:

(14) *Abdullah-ne roṭiyān khā-ī*

Abdullah-ERG breads.F.PL eat-PFV.F.PL

'Abdullah ate breads' → only the object controls the morphology

Unlike Multiple Agree in languages such as Basque or Georgian where both subject and object ϕ -features can be reflected simultaneously in the verbal morphology. This is not expected from Hindi-Urdu which instead shows a one-argument outcome as the accessible object determines the form, with the ergative subject leaving no agreement reflex. Thus, Hindi-Urdu shows case-conditioned intervention rather than simultaneous agreement with multiple arguments.

The MPH analysis: The Hindi-Urdu pattern is more naturally derived by allowing Asp[$i\phi$] to search selectively rather than to establish ϕ -relations with all available arguments at once. In the imperfective, the nominative subject is accessible and determines the relevant morphology. Perfective clauses are different: ergative marking places the subject outside the matching relation, leaving the object as the potential source of ϕ -values. If no appropriate object is available, no matching relation is established and the default form surfaces. Contrary to the simultaneous agreement of a

probe with multiple arguments, a case-sensitive matching process in which accessibility determines which argument, if any, can control the morphology emerges in Hindi-Urdu
Warlpiri: Templatic Agreement and Free Word Order

A different but complementary challenge to the Multiple Agree is posed by Warlpiri due to the fixed sequence of subject and object ϕ -features. Unlike configurational languages, its auxiliary complex (Agr) carries a fixed sequence of subject- and object-agreement markers, occupying the fixed familiar second-position slot:

(15)

- a. *Ngarrka-ngku ka-rna-ngku nya-nyi kurdu-ku*
man-ERG PRES-1SG.S-2SG.O see-NPST child-DAT
- b. *Kurdu-ku ngarrka-ngku ka-rna-ngku nya-nyi*
[Same meaning, with free DP order]

It is important to note that, without affecting the order of the agreement morphology, the DPs can change their surface positions. The auxiliary continues to show the same subject > object sequence even if the corresponding DPs occur in different positions. Nor is the auxiliary's second-position placement determined by the surface location of either DP.

This separation between agreement morphology and surface word order is significant. Had Multiple Agree been directly tied to the structural positions of the arguments being probed, there would have been substantial changes in DP order as consequences for the agreement configuration. But they are none. A more plausible analysis is that the auxiliary establishes its agreement relations within the vP domain, where the subject-object hierarchy is represented, and subsequently moves to the second position (Legate 2002). Its movement satisfies [uEPP] through movement of the pivot itself rather than by attracting the associated phrases.

The Warlpiri evidence points to a useful distinction between agreement structure and surface word order. The sequence of subject- and object-agreement markers remains stable even when the corresponding DPs can appear in different positions. The morphology therefore reflects the underlying agreement configuration rather than the order in which the arguments happen to be pronounced. This separation poses a problem for a version of Multiple Agree that directly links probing relations to the surface positions of the relevant DPs.

Conclusion: Multiple Agree offers a compelling account of polypersonal agreement in configurational languages, where a probe can establish relations with more than one structurally accessible goal. Its explanatory reach becomes less clear, however, in cases where access is restricted by case or where agreement structure is separated from surface word order. Hindi-Urdu illustrates the first problem: ergative marking prevents ϕ -matching with the subject even though both the probe and the subject carry [$i\phi$]. Warlpiri illustrates the second, since agreement is established independently of the freely varying positions of the DPs, while the auxiliary moves to satisfy the EPP. The MPH captures both patterns by treating agreement as Bimodal Agree, with case determining accessibility, and by allowing the EPP to be satisfied through language-specific pivot movement rather than a uniform subject-attraction mechanism.

4.5 Theoretical significance of the ϕ -parameterization

Recent minimalist work has produced a proliferation of auxiliary mechanisms: Feature Inheritance (Chomsky 2008), Feature Sharing (Pesetsky & Torrego 2007), Upward Agree (Bjorkman & Zeijlstra 2019), and Failed Agree (Preminger 2014). Together they reveal an unresolved gap in the theory of ϕ -agreement, since each introduces additional machinery to accommodate empirical configurations that a single, undifferentiated

notion of ϕ cannot capture. In effect, these mechanisms already acknowledge that ϕ -features are not uniform objects. They vary in their availability on functional heads, and in whether their values contribute to interpretation at LF.

The present analysis makes this implicit distinction explicit, formalizing it within a two-tiered ϕ -parametric architecture that is validated both statistically (§3.5.1–3.5.4) and against independent corpus data external to the theory (§3.5.5). Once ϕ -availability and ϕ -interpretability are parameterized as distinct choices, the empirical domains that previously required inheritance, sharing, or repair follow from general principles instead.

This reframing supports a view of MPH as a genuinely integrative framework, one that identifies the latent structural dimension earlier frameworks were converging on from different directions. Corroborated by both internal statistical structure and external corpus data, the dual-validated two-tiered ϕ -parameters supply the generalization that was missing, restoring explanatory continuity across the Agree-based literature.

5. Counterarguments and Theoretical Implications

5.1 Addressing Standard Objections

Several theoretical objections could be raised against pivot-based analyses more generally, and the framework proposed here anticipates them. This section shows that the Hierarchical Parametric Organization withstands these objections and offers empirical advantages over competing approaches.

Legate (2008) and Baker (2015) raise a circularity objection in identifying the host of the EPP bundle arguing that identifying such a functional head risks identifying the pivot simply as "the head that licenses the subject". This would make EPP-related behavior trivially follow. However, the MPH framework breaks this circle through its Biphasic Identification Protocol. Following the protocol, the pivot is identified syntactically first, through independent clause-boundary and movement diagnostics, and only then corroborated morphologically via Bybee's (1985) criteria. Only heads that pass at least two diagnostics from each family qualify for morphological testing. This sequence keeps pivot selection replicable and falsifiable across analysts and datasets. The common minimalist assumption that the EPP is universally hosted by T is being contradicted by the cross-linguistic evidence: Aspect serves as pivot in Indo-Aryan languages, Agreement in Bantu, Mood in Romance, fused Aspect-Evidential systems in Balti. The EPP Hosting Maxim derives this distribution without stipulating an *a priori* universal T. This maxim is a refinement of the feature-anchoring principle, not an abandonment of Minimalist architecture: the EPP is universally required but not universally located.

The complete absence of ϕ -features on Balti Asp/Evid or why there exists a correlation between ϕ -interactions and morphological richness across unrelated families pose challenge to the Feature Inheritance (Chomsky 2007, 2008). In the hierarchical parametric organization proposed by the MPH framework, ϕ -availability is a lexical property of the pivot itself instead of being a result of top-down inheritance. Across finite clauses, Feature Inheritance predicts uniform ϕ -behavior; the attested cross-linguistic split in ϕ -availability confirms the parametric account instead.

Preminger's (2014) Flexible Agree predicts valuation failure should not crash derivations. However, it does not predict the systematic coupling of case, EPP, and ϕ -availability found here because, ϕ -valuation never initiates yet EPP and case are obligatorily realized in Balti while, ϕ -valuation is necessary for EPP satisfaction in English. This polarity requires Bimodal Agree, which Flexible Agree lacks a mechanism to derive.

The unreliable morphology objection, raised by Harley & Noyer 1999, misunderstands the role of morphology in the pivot identification process. It needs to be understood that Bybee's criteria only corroborate syntactic findings independently obtained through structural diagnostics including clause-boundary and movement diagnostics, rather than inferring structure from surface form. Balti evidentials lacking φ -agreement yet enforcing EPP and German, Icelandic, Faroese, and Finnish retaining rich agreement without interpretability show the divergence of syntax and morphology (§3.5.2–3.5.3). The MPH framework analysis treats this divergence as crucial evidence, instead of noise to explain away. Thus, morphology plays a diagnostic, not deterministic, role.

It might be argued that by positing two more parameters contributes to the parameter proliferation objection and increases complexity relative to single-parameter accounts of pro-drop or agreement. However, contrary to the objection, the two-level architecture instead reduces total parameterization: traditional accounts require independent parameters for pro-drop availability, agreement richness, expletive requirements, case systems, and EPP satisfaction. MPH derives all five from two binary choices, capturing their tight cross-linguistic clustering (PC1 = 96.60% morphosyntactic variance explained, §3.5.2) and independently confirmed via corpus data external to the theory (§3.5.5). The resulting system obeys Minimalist economy: one locus (the pivot), one feature class (φ -availability with binary interpretability).

The Mixed systems such as Spanish and Turkish which demonstrate combination of φ -agreement and discourse-based pro-drop do not constitute counter examples; instead, these mixed systems display exactly the kind of pattern the framework allows for. In such mixed systems, interpretable φ -features on the pivot can interact with recoverable φ -information supplied by the discourse, giving rise to a hybrid Agree system rather than forcing pro-drop into a single licensing mechanism. English infinitives illustrate a different kind of reduction. Even though subjects can occur without displaying overt φ -agreement, the relevant non-finite clause lacks the fully specified morphosyntactic pivot found in finite clauses; [iCase] can instead remain associated with the higher, matrix head.

A methodological qualification about the role of freezing diagnostic in identifying the morphosyntactic pivot. We have observed across the five core languages having primary judgement data that freezing diagnostics consistently proved less reliable than the other movement diagnostics (§2). Swahili, English, Italian, and Balti uniformly produced relatively weak or inconclusive effects, and freezing had the lowest inter-rater reliability of the six diagnostics ($\kappa = 0.69$; Appendix C). I recommend viewing this result as a limitation of the diagnostic itself instead of being evidence against the identification of the pivot. Extraction patterns vary considerably across languages, and possessor-extraction effects can further obscure freezing judgments. Crucially, however, the analysis does not depend on freezing alone. The nomination rule requires at least two independent movement diagnostics before a head can be identified as the pivot.

The other diagnostics provide considerably stronger and more consistent evidence: coordination ($\kappa = 0.89$), complementizer selection ($\kappa = 0.84$), ellipsis ($\kappa = 0.79$), extraction ($\kappa = 0.77$), and subject position ($\kappa = 0.73$). Taken together, they yield an overall κ of 0.79 (95% CI [0.76, 0.82]; Appendix C). Freezing therefore contributes to the overall assessment without carrying the burden of establishing pivothood by itself. Its weaker performance is reflected in the weighting of the evidence rather than allowed to determine the result.

More broadly, the standard objections do not account for the central cross-linguistic pattern. They may capture individual pieces of the data, but they do not explain why φ -

availability, case licensing, and the availability of pro-drop pattern together across the languages examined. Bimodal Agree derives that connection as part of the architecture instead of treating each correlation as an isolated fact, unrelated to other correlations.

5.2 The Objectivity of the Coding Schema and the PCA Results

A reasonable methodological question is whether the Principal Component Analysis genuinely tests the MPH or simply reproduces assumptions that were built into it from the outset. This concern primarily arises because of the fact that the variables entered into the PCA are motivated by the MPH framework itself. However, theoretical predictions are separated from the empirical measurements by the explicit operationalization of these variables.

First, instead of an impressionistic judgment about whether a language “looks like” a particular MPH type, each PCA variable is tied to an independently stated and reproducible diagnostic. Rich_Agreement is defined in terms of Bybee’s (1985) criterion of at least three contrastive forms; Morph_ProDrop is established through story-initial elicitation, with a threshold of at least 80% speaker acceptance of null subjects; and No_Expletive is tested in context-neutral weather-verb constructions. The same procedures were applied across the language sample. Thus, although the choice of variables follows from the theoretical questions posed by the MPH, the classifications themselves are determined by explicit empirical criteria.

This point is especially important because the PCA variables differ methodologically from the Phase 1 diagnostics used to identify the pivot. The latter involve graded grammaticality judgments and are therefore open to disagreement between coders. For that reason, they were independently double-coded on a 15% subset ($n = 1,200$ items), producing an overall κ of 0.79 (95% CI [0.76, 0.82]; Appendix C). The four variables entered into the PCA, by contrast, are categorical outcomes generated from fixed thresholds. The reproducibility of these results is not dependent on two researchers reaching the same subjective judgment; rather it follows from applying the same procedure to the same evidence. A researcher using Bybee’s criterion of ≥ 3 -contrastive-forms, the acceptance threshold of $\geq 80\%$ for elicited null subjects, and the context-neutral weather-verb test should obtain the same classifications from the same data. Thus, the methodological distinction is maintained deliberately. Since they involve graded judgments, the Phase 1 diagnostics require inter-rater reliability. On the other hand, the Phase 2 variables require procedural reproducibility because their classifications are generated by predetermined quantitative criteria. Treating the two sets of diagnostics in the same way would actually obscure an important difference in their evidential status. The contrast between them reflects different measurement problems, not a difference in methodological standards.

Second, the variables are independently testable because a researcher can independently apply the diagnostic battery provided in Appendix D to a new language and generate a coding profile without any commitment to the MPH framework or the prior knowledge of its predictions. The MPH framework’s capacity to withstand blind, objective application is evident from its application over seventy-five outgroup languages and UD corpus data of over 2 million sentences.

Third, the analysis is corroborated by structure it did not build in. It has now been decisively proved on two independent grounds rather than one. The internal statistical evidence needs to be stated carefully here. PC1 explains 96.60% of variance across the four hand-coded diagnostics (§3.5.2), and the polychoric correlation matrix (§3.5.3) reaches its estimation boundary, with all pairwise coefficients at approximately 1.000. This reflects a genuine empirical fact: once corrected, the sample contains essentially no

counterexamples on three of the four diagnostics. A sceptical reader might reasonably wonder whether concentration this high itself is evidence of circularity. It is not, and the reason does not depend on PC1's magnitude at all: the diagnostics were tested against an evidence source that had no part in generating them. *Phi_Availability* values derived mechanically from raw Universal Dependencies treebank morphology — a corpus built by independent annotators with no relationship to MPH — recover the hand-coded classifications at 95–97.5% accuracy (§3.5.5), using a threshold fixed before the test was run. This is the decisive non-circularity evidence: not an internal property of the PCA (which, on a highly deterministic four-variable system, will always show high PC1 regardless of whether the underlying construct is real), but an external test the theory could have failed and did not.

The internal analysis does retain real evidential value, though for a different reason than originally claimed. PC2 is small, at 3.40%, but it is not noise; it isolates exactly one theoretically motivated distinction. *Rich_Agreement* contributes 76.83% of PC2's variance, separating Type 2a, with impoverished morphology (English, Dutch, French), from Type 2b, which retains rich morphology despite an uninterpretable status (German, Icelandic, Faroese, Finnish). Theoretically most interesting empirical finding of this study (§3.2.2, §3.5.2) is the dissociation between agreement richness and interpretability. If the circularity were real and four variables were mere restatements of one theoretical claim, this structured, independently interpretable residual is not what would result. Homogeneous noise would.

It is important to precisely state the falsifiable claim. As established in §1.9.2, the theory stands falsified if we come across a language in which *Phi_Availability*, *Morph_ProDrop*, or *No_Expletive* diverge. As we have observed, this deterministic trio holds at $r=1.000$ across all 82 sample languages, with no exception. In case $r=1.000$ does not hold, the theory is falsified. As we have shown, *Rich_Agreement* creates distinction within type instead of among types whereas the remaining diagnostics create typological instead of within-type distinctions. Therefore, *Rich_Agreement* is explicitly excluded from this inviolable core; and a language showing rich paradigm alongside the explicit requirement for expletives does not constitute a counterexample but an instance of the predicted Type 2b pattern, independently confirmed in four languages via both internal coding and external UD corpus data (§3.5.5). Stating the falsifiable claim in precise terms here is a strength, not a hedge because it specifies in advance exactly which combinations are forbidden and which are an expected, independently-motivated exception with both halves of that specification being borne out empirically.

The decisive test of non-circularity, restated: could a researcher blind to MPH generate the same coding, and would an entirely independent data source agree with it? Both are now answered affirmatively with evidence: the diagnostic battery is fully specified and quantitative (Appendix E), application to seventy-five outgroup languages succeeded without foreknowledge of predictions, and the resulting codings were independently corroborated by Universal Dependencies treebank data the researcher did not construct (§3.5.5). This is a stronger standard of non-circularity than internal statistical structure alone can provide, and the study meets it.

5.3 Typological Predictions and Empirical Scope

The two-level parametric architecture generates specific testable predictions about possible and impossible language types, creating falsifiable boundaries on cross-linguistic variation.

Predicted impossibilities:

- Languages should not show morphologically licensed pro-drop while also requiring obligatory expletive insertion, or vice versa, since these diagnose interpretable and uninterpretable ϕ respectively and cannot co-occur on the same pivot (§3.5.3, §3.8). Rich person-number agreement morphology is a different matter and is explicitly not predicted to track this distinction perfectly. German, Icelandic, Faroese, and Finnish show rich agreement alongside expletive-obligatoriness and blocked pro-drop, a stable, independently confirmed pattern (Type 2b; §3.2.2, §3.5.5) that reflects agreement's capacity to survive diachronically after a pivot's ϕ -features have already become uninterpretable (§5.5). Far from a rare or transitional exception, this is a predictable consequence of the theory's own claim that Rich_Agreement is dissociable from the other three diagnostics.
- Languages should not lack person-number/gender morphology entirely while showing morphologically mediated pivot-based pro-drop, since morphological licensing requires ϕ -information recoverable from the pivot.
- Languages should not show discourse-insensitive pro-drop without agreement morphology, since only pivot-mediated pro-drop through interpretable ϕ -features permits pro-drop independent of discourse context.

Predicted property clustering:

All ϕ -unavailable systems should uniformly lack person-number/gender paradigms on pivots, show aspectual or evidential morphology as clause-defining, exhibit aspect-conditioned (not ϕ -conditioned) case alternations, permit discourse-mediated pro-drop with person-hierarchy effects, and show topic-structure sensitivity in null-subject licensing.

All ϕ -available systems with interpretable features should show rich agreement paradigms, permit pro-drop without discourse restrictions, lack expletive requirements, and show ϕ -interacting case systems.

All ϕ -available systems with uninterpretable features should require overt subjects or expletives, block pro-drop, and tie case licensing to ϕ -valuation — **but agreement richness within this type is not uniform: Type 2a (English, Dutch, French) shows impoverished agreement; Type 2b (German, Icelandic, Faroese, Finnish) retains rich agreement despite uninterpretable status.** This split is itself a confirmed prediction, not a complication: it follows directly from Rich_Agreement's independent diachronic behavior (§3.5.6, §5.5).

Extensibility to other systems: In the MPH, as pivothood is a formal construct and is empirically defined through replicable diagnostics, it is natural for the framework to extend to other alignment systems, e.g., Philippine-type voice systems, Agent-Focus (A'-driven) systems, Quechuan switch-reference. Each new application requires the same Biphasic protocol, following the same procedure of identifying the functional head which passes clause-boundary and movement diagnostics, and, then, is corroborated through morphological diagnostics. The investigation of understudied languages through MPH is expected to reveal additional pivot types, potentially including voice-based pivots, evidential-only pivots without aspectual specification, or mood-aspect fusions distinct from the Italian pattern.

Bounded typological diversity: The framework predicts bounded, not unlimited, typological diversity: three surface types (0.1, 0.2, 1.0), with Type 2 internally structured into 2a/2b by agreement richness rather than constituting a fourth macro-type. Mixed or transitional systems like Spanish and Turkish represent parameter interaction or diachronic drift in ϕ -availability, not additional types. Surface changes in

agreement richness are predicted to occur more frequently than changes in fundamental clause-anchoring mechanism, since shifts in φ -interpretability within φ -available systems require less radical reanalysis than shifts between φ -available and φ -less systems — a prediction for which Type 2b now constitutes direct, attested confirmation (§5.5), not merely a theoretical possibility.

5.4 Acquisition and Processing Implications

Learnability follows directly from surface-visible morphological cues, supporting minimalist acquisition models emphasizing distributional learning from primary linguistic data.

Children acquiring φ -available systems with interpretable features should master agreement paradigms early, produce appropriate pro-drop from the earliest multiword stage, and show no extended period of expletive-omission errors, since expletives are unnecessary. Swahili acquisition studies (Deen 2003, 2006; Demuth 2003) confirm this: subject prefixes appear by eighteen months, with near-errorless agreement mastery by age two and null subjects produced appropriately from the start.

Children acquiring φ -available systems with uninterpretable features should produce null-subject errors during an Optional Infinitive stage before recognizing overt subjects are required and show extended expletive-omission errors before mastering that expletives satisfy EPP requirements. English acquisition confirms this: null subjects during the Optional Infinitive stage around ages two to three, late expletive-insertion acquisition around ages three to four (Pérez-Leroux & Wexler 1999), with extended omission errors before convergence on obligatory overt subjects by ages four to five.

Critically, this delay is predicted to track interpretability rather than paradigm richness (§3.8): German — despite rich adult agreement — shows a comparable Optional Infinitive stage to English, distinguishing this account from one where delay simply reflects paradigm poverty.

Children acquiring φ -less systems are predicted to master aspectual and evidential morphology early, since it is the clause-defining system, show discourse-appropriate pro-drop from early stages, and potentially overgeneralize pro-drop to discourse-inappropriate contexts before fully acquiring pragmatic licensing conditions. Acquisition data for Balti and related Tibetan languages remain limited; these predictions are testable through future fieldwork and provide clear falsification criteria.

Surface visibility and early parameter setting supports Wexler's (1998) Very Early Parameter Setting hypothesis: rich paradigms provide direct positive evidence for interpretable φ -features; sparse paradigms combined with obligatory expletives provide direct positive evidence for uninterpretable φ -features **in Type 2a systems specifically** — for Type 2b, the relevant positive evidence is expletive-obligatoriness and blocked pro-drop *despite* rich paradigms, since paradigm richness alone would misleadingly suggest interpretability. Absence of φ -morphology combined with discourse-conditioned null subjects provides direct positive evidence for φ -less pivots. Convergence is predicted by ages two to three across all systems.

5.5 Diachronic Trajectories and Language Change

The asymmetric patterns in language change are predicted by the hierarchical parametric organization of the locus of the EPP. Within φ -available systems, shifts in φ -interpretability may occur through paradigm erosion or enrichment without fundamentally restructuring clause architecture, and this is more than a theoretical possibility — it is directly attested in the Type 2b cluster.

As we observed in the case of German, Icelandic, Faroese, and Finnish, they retain rich verb paradigm, productive agreement morphology which they have inherited from their

earlier interpretable stages. Their pivots have independently become uninterpretable, as confirmed by expletive-obligatoriness and blocked pro-drop (§3.2.2, §3.5.5) but they yet retain the rich paradigm. Thus, morphology and interpretability become decoupled mid-transition, retaining agreement which outlasts interpretability loss. The predicted asymmetry becomes concrete in this pattern: agreement richness is conservative, and changes more slowly than the syntactic property it once reliably signalled.

Conversely, analogical extension of paradigms could shift systems from uninterpretable toward interpretable ϕ -features, potentially enabling pro-drop acquisition as morphological licensing becomes possible, though no case in the current sample is analyzed as undergoing this direction of change.

Shifts between ϕ -available and ϕ -less systems require more radical reanalysis, potentially driven by reinterpretation of aspectual or evidential morphology. When aspectual markers become obligatory through grammaticalization, learners might reanalyse them as clause-defining rather than optional tense modifiers, potentially triggering a shift from ϕ -based to aspect-based clause anchoring if agreement paradigms simultaneously erode. Conversely, grammaticalizing person-number markers from pronouns or auxiliaries could trigger the reverse shift, from aspect-based to ϕ -based anchoring.

The Hindi-Urdu case: the split between perfective (ergative case, object agreement) and imperfective (nominative case, subject agreement) contexts could reflect either stable aspectual conditioning (as in Georgian) or an intermediate diachronic stage. Historical evidence from Indo-Aryan (Butt 2001; Verbeke & De Cuyper 2009) shows ergative patterns arising through reanalysis of perfective participles, suggesting Hindi-Urdu represents a stable split-ergative system rather than a transitional state, predicting continued coexistence of both patterns.

The Italian case: The morphological fusion of mood and tense in Italian poses an interesting case. This fusion seems to indicate a particular stage where the syntactic centrality of the morphosyntactic pivot is fixed but morphology is yet in the process of revealing the morphological prominence of the pivot. The process of diachronic change has reached a stage where the structural centrality of mood is fully established though morphology is yet extensively fused. The clitic placement's sensitivity to mood is especially revealing as the relevant syntactic organization remains visible even where the morphology no longer transparently reflects the morphological prominence of the morphosyntactic pivot. This raises the possibility that Italian-type systems can undergo gradual reorganization, with the grammar moving either toward renewed morphological differentiation or, at the other extreme, toward more extensive fusion.

5.6 Connections to Broader Theoretical Debates

The MPH also bears on several longstanding questions within the Minimalist Program. One such question is particularly how much of a clause structure should be shared across languages and how much freedom should be allowed for cross-linguistic variation. Cartographic approaches (Rizzi 1997, Cinque 1999, and Belletti 2004), derive much of this variation from relatively elaborate universal hierarchies in which functional projections occupy fixed positions. The non-cartographic approaches, on the other hand, allow considerably more flexibility, treating functional structure as something that can differ substantially from one language to another.

The MPH takes a position between these two extremes. Neither does it dispense with universal structural requirements, nor does it assume in advance that T, or any other particular functional head, must be the universal EPP host in every language. However, the grammar must provide a head capable of hosting the EPP bundle and licensing case

to the subject DP; what is left open is which functional head takes on that role. That choice is determined parametrically across languages. This provides a more flexible clause architecture in which projections may be absent, reordered, or organized differently across languages. Despite not assuming T as the universal EPP host, the analysis still remains minimalist without giving up the Minimalist view that cross-linguistic variation is constrained by a universal inventory of features and operations.

On morphology and syntax: The relationship between morphological realization and syntactic structure raises a related question: does the morphology tell us which head functions as the structural pivot, or can morphological exponence and syntactic organization diverge? The MPH does not take either possibility to the extreme.

Morphological prominence is an important piece of evidence for identifying the pivot, but it is not sufficient on its own to establish pivothood. The Biphasic Diagnostic Protocol therefore gives priority to independent syntactic evidence. Clause-boundary diagnostics and movement tests establish the structural position of the pivot, while morphological prominence provides an additional, independently motivated source of support.

Bimodal Agree and syntactic operations: The fact that case valuation can proceed without any φ -interaction in languages whose pivots lack φ -features has broader consequences for the theory of agreement. It suggests that the featural makeup of functional heads is considerably more variable across languages than analyses built around a universally φ -bearing T would allow. The variation may therefore extend beyond the lexical inventory of features themselves. Some aspects of the syntactic operations that make use of those features may also be parametrically determined, rather than being entirely universal with cross-linguistic differences confined to which features happen to be present.

Syntax-pragmatics interface: The contrast between pivot-mediated and discourse-mediated pro-drop also raises a broader question about how much of subject recoverability is determined by narrow syntax and how much is left to the interfaces. The two mechanisms suggest that languages can arrive at the same surface outcome—an unexpressed subject—by relying on different resources. Where the pivot carries interpretable φ -features, morphological agreement can identify the null subject without depending on the discourse. In φ -less systems, that information is not supplied by the syntax; recoverability instead depends on the discourse context. The prediction is therefore fairly clear: interpretable- φ systems should allow pro-drop independently of discourse conditions, whereas languages with φ -less pivots should show greater sensitivity to discourse. This provides a concrete way of testing where the boundary between grammatical licensing and pragmatic interpretation is drawn across languages.

Conclusion

The present study aimed to establish that, instead of being the universal projectional property of T, EPP is a derivative property of the morphosyntactic pivot which is identified through independent syntactic and morphological diagnostics. Every language must have a functional head serving as the morphosyntactic pivot, but which functional head serves as the morphosyntactic pivot varies from language to language. The status of the φ -availability on the pivot crucially determines the mode of Agree which further determines the cross-linguistic distribution of subjects, case, agreement, expletives, and pro-drop. Under this unified perspective, language-specific stipulations are replaced with a two-level parametric architecture which creates syntactic dependencies in clause structure across languages.

As the structural and morphological diagnostics point to the same functional head as the morphosyntactic pivot, it restores the symmetry between syntax and morphology. The two-stage identification protocol, which can be independently replicated, offers a pathway from surface diagnostics to theoretical generalization. The syntactic identification of the pivot and its morphological corroboration significantly reduce the risk of circularity that has plagued comparative syntax. Long assumed to be universally hosted by T, EPP emerges as a universal requirement whose host may vary from language to language.

This claim is now supported by two independent lines of evidence rather than one. The Principal Component Analysis provides robust, quantitative support internal to the coded data: the dominant dimension of variation (PC1 = 96.60%) cleanly separates languages by a single latent parameter — a language's φ -state, whose interpretable- φ setting ([i φ]) is one of its four convergent reflexes. Independently, corpus-derived φ -availability values, computed mechanically from raw Universal Dependencies treebank morphology with no reference to MPH, recover the hand-coded classifications at 95–97.5% accuracy (§3.5.5). The theory's central claim is thus corroborated both by structure internal to its own data and by an external test it could have failed.

Critically, the strongest confirmation of the central theoretical claim comes not from proximity on PC1 but from the behavioural pattern itself (§1.6.3, P3–P4): Balti, a [n φ] pivot entirely lacking φ -features, and the [i φ] languages, whose pivots bear fully interpretable φ -features, converge on identical surface behavior — both permit null subjects and disallow expletives — despite sitting at opposite ends of the φ -Availability parameter. English, a [u φ] pivot, is the outlier: possessing a φ -probe that neither the φ -less Balti nor the φ -interpretable [i φ] languages need in the same way, English alone blocks null subjects and requires expletives. This U-shaped pattern is not predicted by morphological richness — Balti's extensive aspectual-evidential paradigm is more elaborate than English's impoverished agreement, yet English is the behavioural outlier, not Balti — nor by φ -Availability alone, which would place [u φ] English between [n φ] Balti and the [i φ] languages rather than apart from both. It is exactly the signature predicted by φ -Interpretability specifically: the pivotal syntactic distinction is not between systems with 'more' or 'less' morphological structure, but between systems where the pivot's φ -features are semantically active at LF and those where they are not — including the null case where no φ -features are present to begin with.

The secondary dimension (PC2 = 3.40%) then reveals finer structure that PC1 alone does not capture. Rich_Agreement dominates PC2's loading, and its effect is concentrated on isolating Type 2b (German, Icelandic, Faroese, Finnish) — languages whose retained rich agreement sets them apart from every other group, including Type 3, despite Type 2b sharing Type 3's morphological richness and sharing Type 1/Type 2a's uninterpretable or absent φ -status on the primary dimension. PC2 is thus not a smooth three-way subdivision of the "0" cluster. It is the residual signature of the dissociation between agreement-richness and interpretability which this study identifies as its central empirical discovery. This geometric description is confirmed directly by the biplot for the 82-language sample (Figure 8): Rich_Agreement contributes 76.83% to PC2, versus 7.72% for each of the other three variables (Table 7), and Type 2b sits well apart from every other group on PC2, exactly as predicted.

This dual-evidence structure also addresses a standard objection to typological PCA studies before it is raised: that apparent clustering reflects genealogical or areal bias rather than a genuine cross-linguistic parameter. A supplementary analysis splitting the sample by Indo-European membership (N=41 each) shows the primary dimension

holds independently in both subsamples (Indo-European PC1=92.05%; non-Indo-European PC1=98.51%) — if anything, marginally *stronger* outside the genealogically dominant family. The pattern indicates that these results are not an artifact of the sample's Indo-European weighting.

The economy is achieved by deriving multiple phenomena including subject licensing, case systems, agreement patterns, expletives, and pro-drop from the same single axis. The hierarchically organized locus of the EPP reduces the typological space. Bounded typology generated by a set of two binary parameters locked in a logical-dependency relation predicts possible and impossible language types while accommodating documented diversity across language families. The surface visibility of morphological cues ensures learnability, whereas the hierarchical organization of the locus of the EPP predicts asymmetric patterns in diachronic change. These patterns, as observable in Type 2b languages, now constitute direct attested confirmation, not merely a theoretical possibility (§5.5). This dual-validated model — statistically robust and independently corroborated by external corpus data — offers a falsifiable model of clause structure in the classic Popperian sense (§1.9.2): its central predictions could have been contradicted by either line of the evidence and were not.

This work's contribution to parametric theory goes beyond the empirical findings about pivot variation: it demonstrates that a parameter can exhibit logical dependency, with one parameter's definition conditional on another's value. Through this dependency, the φ -Availability/ φ -Interpretability system generates three surface types rather than $2^2 = 4$, with the impossible fourth type ruled out by logical structure rather than by stipulation.

Future research should focus on extending the diagnostic protocols to understudied language families, test predictions about acquisition trajectories in φ -less systems and examine whether additional pivot types exist in languages with different morphological profiles. An attempt was made to extend the corpus-based validation of §3.5.5 to Rich_Agreement, Morph_ProDrop, and No_Expletive using UD's available proxies (agreement-feature richness, null-subject rate, and expl-relation frequency respectively), but this effort did not succeed: unlike verbal morphology, which UD's FEATS annotation records in a comparatively direct and theory-neutral way, pro-drop and expletive status are structural properties established through syntactic diagnostics (coordination, ellipsis, movement) that a raw count of surface dependency relations cannot reliably approximate — the resulting proxies conflate genuine morphological licensing with topic-drop, imperatives, and treebank-specific annotation conventions, and their inclusion in a multi-factor model degraded rather than improved classification accuracy under honest cross-validation. This evidence suggests that, instead of being achievable through better feature engineering on the existing resource, corpus-based validation of the full four-variable bundle may require treebanks with richer syntactic annotation than UD's dependency scheme currently provides. Corpus studies tracking diachronic shifts in φ -availability would further illuminate the mechanisms of parametric change directly attested in Type 2b. The primary criterion for evaluating the theory should therefore be the reproducibility of its classifications across independently investigated languages rather than agreement with any individual descriptive tradition. Ultimately, the long-term value of the Morphosyntactic Pivot Theory will be determined by the cumulative results of independent researchers applying the documented protocol across the widest possible range of the world's languages.

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